

A Structure-Constrained Neural Surrogate for the Dirichlet–Neumann Operator in Nonlinear Water-Wave Simulation

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Abstract

We study neural approximation of the finite-depth Dirichlet–Neumann operator for the two-dimensional irrotational water-wave problem with one horizontal coordinate. The central difficulty is that a small error in one operator evaluation need not remain small after repeated use inside a Hamiltonian time integrator. Our surrogate retains the exact zeroth- and first-order Craig–Sulem terms and represents only the second- and higher-order contribution by a self-adjoint neural residual. Its training objective combines relative L^2 data misfit with Fourier-mode, local translation-tangent, and Hadamard shape-derivative consistency terms. Reference trajectories use a double-precision order-six Craig–Sulem expansion, padded nonlinear products, and an integrating-factor two-stage Gauss–Legendre method. The reported checkpoints use the historical v9 snapshot archive. The parameterized, case-split replacement dataset specified in this paper is complete and source-authenticated, but no reported checkpoint was trained on it.

An unguarded 320-trajectory test of an earlier model with the same structural decomposition contains no nonfinite surrogate rollout, although two finite Tanaka trajectories exceed unit relative surface error. Under the stated soliton-selective guarded protocol, the selected model remains finite in all 320 registered rollouts drawn from ten panels representing five initial-wave families. Among the 305 cases with valid reference trajectories, the terminal relative surface error has median 0.00285 and 95th percentile 0.04769. Removing only the Fourier-mode term improves one-step validation error but worsens the pooled terminal 95th percentile by 31% and the mean by 45%. These results show that one-step value accuracy alone does not predict the long-time error tail. They also leave a clear limitation: the strongest current accuracy result is conditional on the declared guard, and its worst remaining error is a finite Tanaka error.

1 Problem introduction

We consider two-dimensional, inviscid, incompressible, irrotational gravity waves over a horizontal bed. The free surface is represented by $y = \eta(x, t)$, the undisturbed depth is $h > 0$, and $\xi(x, t) = \phi(x, \eta(x, t), t)$ is the trace of the velocity potential on the free surface. Gravity acts with acceleration $g > 0$. The horizontal coordinate is periodic, $x \in \mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$, with period $L > 0$. The potential solves

$$\Delta\phi = 0 \quad -h < y < \eta(x, t), \quad (1)$$

$$\partial_y\phi = 0 \quad y = -h, \quad (2)$$

$$\phi = \xi \quad y = \eta(x, t). \quad (3)$$

The Dirichlet–Neumann operator (DNO) maps the surface potential to the unnormalized upward normal derivative,

$$\mathcal{G}(\eta; h)\xi = (\partial_y\phi - \eta_x\partial_x\phi)_{y=\eta(x, t)}. \quad (4)$$

Eliminating the bulk potential gives Zakharov’s canonical surface system [18, 4]:

$$\eta_t = \mathcal{G}(\eta; h)\xi, \quad (5)$$

$$\xi_t = -g\eta - \frac{1}{2}\xi_x^2 + \frac{(\mathcal{G}(\eta; h)\xi + \eta_x\xi_x)^2}{2(1 + \eta_x^2)}. \quad (6)$$

The nonlocal elliptic solve is now isolated in $\mathcal{G}(\eta; h)\xi$, but it must be evaluated repeatedly at the internal stages of the time integrator. This is the dominant cost of the reference solver and is therefore a natural target for operator learning.

The system is Hamiltonian in the canonical variables (η, ξ) , with

$$H(\eta, \xi) = \frac{1}{2} \int_{\mathbb{T}_L} (g\eta^2 + \xi\mathcal{G}(\eta; h)\xi) \, dx, \quad \eta_t = \frac{\delta H}{\delta \xi}, \quad \xi_t = -\frac{\delta H}{\delta \eta}. \quad (7)$$

This structure explains why an accurate static approximation $\mathcal{G}_\theta(\eta; h)\xi \approx \mathcal{G}(\eta; h)\xi$ is not automatically an accurate dynamical model. Varying the kinetic-energy term in (7) with respect to η introduces the surface-shape derivative $D_\eta\mathcal{G}$. If $\mathcal{G}_\theta$ is substituted into both (5) and the geometric expression (6), the resulting equations arise from one learned Hamiltonian only when $\mathcal{G}_\theta$ obeys the corresponding shape-derivative identity.

Our computational goal is consequently stronger than one-step regression: construct a surrogate that

1. is linear and self-adjoint in ξ for fixed (η, h) ;
2. reproduces the exact linear dispersion relation and the leading nonlinear Craig–Sulem correction;
3. is accurate over several wave families and depths; and
4. remains stable when evaluated thousands of times on states generated by its own rollout.

The main contribution is a Craig–Sulem-structured DNO (CS-DNO) whose exact backbone is $\mathcal{G}_0 + \mathcal{G}_1$ and whose learned residual is constrained to be $O(\eta^2)$. The training objective combines relative L^2 regression with an active-mode, phase-space-normalized complex loss, a localized translation-tangent loss, and a randomized finite-secant version of the Hadamard identity. Once a harmonic or sideband is reference-active, the modal normalization gives its relative complex error a weight comparable to that of a stronger carrier. The tangent term separately penalizes the local error component associated with translation, hence with propagation speed, on solitary-wave sources; the Hadamard term constrains the response to changes in surface shape. The design is motivated by a frame-by-frame failure analysis: the unstable models are accurate on exact trajectory states but respond too strongly to small perturbations transverse to the reference trajectory, creating same-sign sidebands in modes $k = 32, \dots, 128$. The nonlinear equations and implicit stage iteration amplify this contamination but do not initiate it.

Throughout, the principal test regimes are (i) Tanaka solitary waves and their interactions, which stress finite-depth nonlinearity and localized spectra, and (ii) Benjamin–Feir-modulated Stokes waves, which stress sideband growth and long-time phase accuracy. Unless stated otherwise, computations use $L = 2\pi$, $N = 1024$, $g = 1$, and a retained Fourier subspace $|k| \leq 128$.

2 Numerical solver: Craig–Sulem expansion and Gauss–Legendre integration

2.1 Reference Dirichlet–Neumann operator

For a horizontal bed and a free surface sufficiently close to the flat state, the DNO is analytic in the surface elevation and admits the Craig–Sulem expansion [4, 16]:

$$\mathcal{G}(\eta; h) = \sum_{m=0}^{\infty} \mathcal{G}_m(\eta; h), \quad \mathcal{G}_m(\lambda\eta; h) = \lambda^m \mathcal{G}_m(\eta; h). \quad (8)$$

Writing $D = -i\partial_x$, the first two terms are

$$\mathcal{G}_0(h) = |D| \tanh(h|D|), \quad (9)$$

$$\mathcal{G}_1(\eta; h)\xi = -\mathcal{G}_0(h)(\eta\mathcal{G}_0(h)\xi) - \partial_x(\eta\partial_x\xi). \quad (10)$$

The historical rollout-derived labels and truth trajectories, as well as every target in the forward dataset design, use the recursion through order $M = 6$. Products inside this expansion are computed by zero-padding Fourier coefficients by a factor of eight, multiplying on the enlarged physical grid, and truncating back to the base grid. Forward-dataset targets are projected onto $|k| \leq 128$ and have zero mean. As documented in [section 3](#), three historical static sources retain unprojected stored labels and are not retroactively described by the forward target convention. These numerical choices are conservative because the order- M recursion contains powers of the differentiation multiplier and can magnify unresolved high-frequency roundoff as $|k|^M$.

The nonlinear expression in (6) is also dealiased. Given $q = \mathcal{G}(\eta; h)\xi$, we evaluate

$$\mathcal{N}_\xi(\eta, \xi, q) = -\frac{1}{2}\xi_x^2 + \frac{1}{2} \frac{(q + \eta_x \xi_x)^2}{1 + \eta_x^2} \quad (11)$$

on a grid with $2N$ points and then spectrally truncate to N points. This implements the equations-of-motion dealiasing advocated by Xu and Guyenne [16]. Our diagnostics show that dealiasing removes a terminal amplifier but does not by itself cure a learned-operator instability: for failing states it delays the first nonfinite value only slightly because the spectral cascade is already established before aliasing becomes large.

2.2 Integrating-factor splitting

We separate the exactly solvable linear gravity-wave system from its nonlinear remainder. With $u = (\eta, \xi)$, define

$$\mathcal{N}(u) = \begin{bmatrix} (\mathcal{G}(\eta; h) - \mathcal{G}_0(h))\xi \\ \mathcal{N}_\xi(\eta, \xi, \mathcal{G}(\eta; h)\xi) \end{bmatrix}. \quad (12)$$

In Fourier space, for each nonzero mode,

$$\frac{d}{dt} \begin{bmatrix} \widehat{\eta}_k \\ \widehat{\xi}_k \end{bmatrix} = L_k \begin{bmatrix} \widehat{\eta}_k \\ \widehat{\xi}_k \end{bmatrix} + \widehat{\mathcal{N}}_k, \quad L_k = \begin{bmatrix} 0 & g_0(k; h) \\ -g & 0 \end{bmatrix}, \quad g_0(k; h) = |k| \tanh(h|k|). \quad (13)$$

Since $L_k^2 = -\omega_k^2 I$ with $\omega_k^2 = gg_0(k; h)$ and I the 2×2 identity, the matrix exponential is available exactly for any time increment τ :

$$e^{\tau L_k} = I \cos(\omega_k \tau) + \frac{L_k}{\omega_k} \sin(\omega_k \tau). \quad (14)$$

With $v(t) = e^{-tL}u(t)$, the transformed equation is

$$v_t = F(t, v) := e^{-tL}\mathcal{N}(e^{tL}v). \quad (15)$$

This removes the stiff linear oscillation from the nonlinear stage equations and gives exact linear dispersion independently of the step size.

2.3 Two-stage Gauss–Legendre method

We apply the two-stage Gauss–Legendre Runge–Kutta method to (15). It is fourth-order accurate and symplectic for a Hamiltonian vector field. Its Butcher coefficients are

$$c_{1,2} = \frac{1}{2} \mp \frac{\sqrt{3}}{6}, \quad A = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \end{bmatrix}, \quad b_1 = b_2 = \frac{1}{2}. \quad (16)$$

For a step from t_n to $t_{n+1} = t_n + \Delta t$, with $v_n \approx v(t_n)$ and a_{ij} the entries of A , the stages solve

$$Y_1 = v_n + \Delta t (a_{11}F(t_n + c_1\Delta t, Y_1) + a_{12}F(t_n + c_2\Delta t, Y_2)), \quad (17)$$

$$Y_2 = v_n + \Delta t (a_{21}F(t_n + c_1\Delta t, Y_1) + a_{22}F(t_n + c_2\Delta t, Y_2)), \quad (18)$$

Define $F_i = F(t_n + c_i\Delta t, Y_i)$ for $i = 1, 2$. The update is

$$v_{n+1} = v_n + \frac{\Delta t}{2} (F_1 + F_2). \quad (19)$$

The historical v9 trajectories and the forward Tanaka and Benjamin–Feir contracts use at most four simultaneous Picard updates for (18). The forward JONSWAP/TMA contract uses at most five; this extra update was fixed together with its larger internal grid and does not change the convergence test. The forward dataset declares a stage converged only when

$$\max_{\substack{i \in \{1,2\} \\ z \in \{\eta, \xi\}}} \frac{\|\Phi_i^z - Y_i^z\|_{\ell^2}}{\max\{\|\Phi_i^z\|_{\ell^2}, \|Y_i^z\|_{\ell^2}, \text{tiny}\}} \leq 10^{-8}, \quad (20)$$

where Φ_i is the right-hand side of the i -th stage equation and tiny is the smallest positive normal float64 number. The superscript $z \in \{\eta, \xi\}$ labels the two component arrays, and $\|\cdot\|_{\ell^2}$ is their discrete norm. The implementation evaluates this norm on Fourier coefficients; Parseval’s identity gives the same ratios as the physical-grid norm. Step sizes and output cadences are specified with their corresponding dataset protocols in section 3. The nonlinear right-hand side is projected after its dealiased evaluation; after each accepted step, both state fields are projected to the same resolved subspace and the mean of ξ is set to zero.

Symplecticity alone does not protect a surrogate rollout. It applies to the Hamiltonian vector field actually supplied to the method. If $\eta_t = \mathcal{G}_\theta(\eta; h)\xi$ is combined with a ξ_t that is not the variation of the same learned kinetic energy, then the vector field is noncanonical and the usual Hamiltonian-structure argument for Gauss–Legendre integration no longer applies. Moreover, once the learned tangent gain becomes sufficiently large, the Picard map in (18) ceases to be contractive. In our failure traces this is the terminal event, not the initial source of the error.

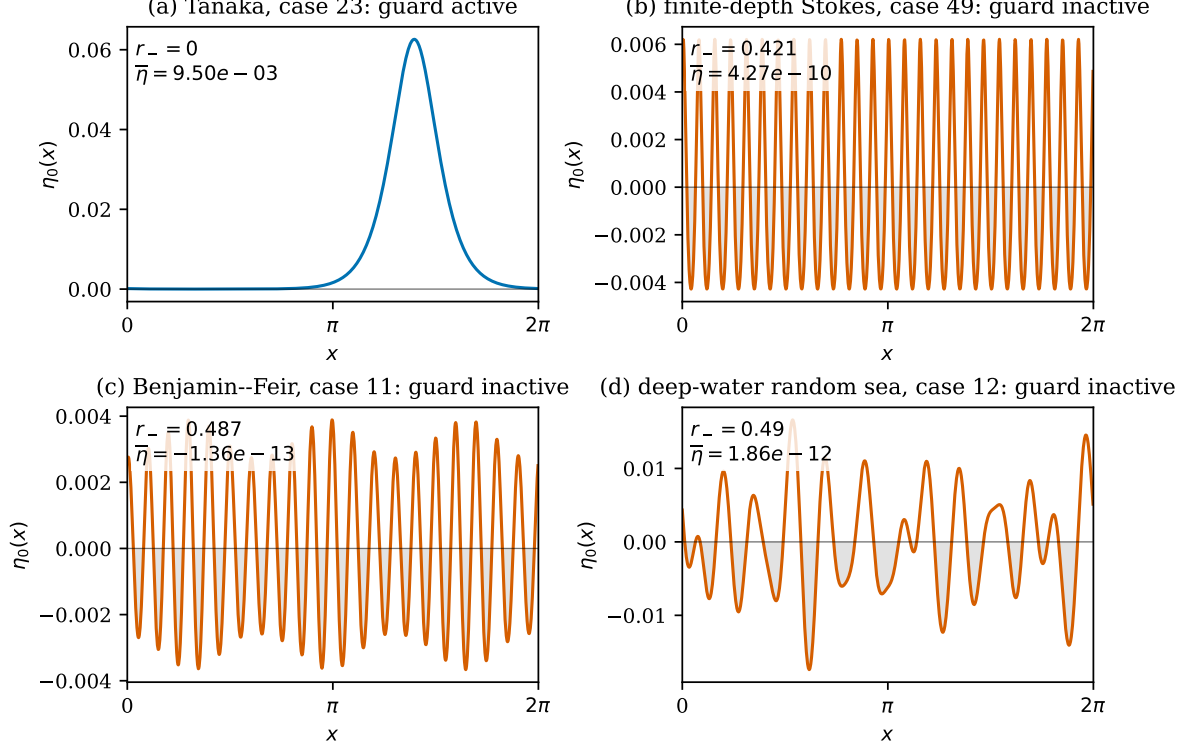


Figure 1: Examples of the initial-condition selector in (22). Gray shading marks the negative part of each surface profile. The Tanaka profile has negative-energy fraction below 10^{-3} and positive mean, so the guard is active. For the three non-Tanaka profiles the negative-energy fraction exceeds the threshold, so the guard is inactive. All four cases remain in the rollout evaluation.

2.4 Soliton-selective spectral-envelope guard

The structured operator removes nonfinite rollouts, but one close two-soliton interaction still develops a resolved spectral cascade. We introduce a conditional stabilization for this failure mode. It is deliberately distinct from the fixed projection used by the reference solver: the guard is exactly inactive whenever the profile-based test below fails, and is nearly the identity on a trajectory for which it is active until the initially empty production band grows abnormally.

First, the initial condition is classified using only its surface profile. Let

$$r_-(\eta_0) = \frac{\|\min(\eta_0, 0)\|_2^2}{\|\eta_0\|_2^2}. \quad (21)$$

Here the minimum is taken pointwise and $\|\cdot\|_2$ is the discrete spatial norm. The guard is active only when

$$s_0 = \mathbf{1}\{r_-(\eta_0) < 10^{-3}\} \mathbf{1}\{\text{mean}(\eta_0) > 0\} = 1. \quad (22)$$

Here $\text{mean}(\eta_0) = N^{-1} \sum_{j=0}^{N-1} \eta_0(x_j)$. This is a profile-based activation test, not a wave-family label. In the evaluation dataset all 64 Tanaka initial conditions satisfy $r_- \leq 2.11 \times 10^{-4}$, whereas the minimum over the inspected non-Tanaka conditions is 0.257.

The selector does not estimate oscillation, localization, or family identity. It tests only the negative-energy fraction and the mean in (22); the examples in figure 1 illustrate the resulting guard state.

For the unnormalized discrete Fourier transform $\hat{\eta}_k$, define the production-band amplitude

$$A(t) = \frac{1}{N} \left(\frac{1}{N} \sum_k \mathbf{1}_{\{64 \leq |k| < 128\}} |\hat{\eta}_k(t)|^2 \right)^{1/2}, \quad A_{\text{tr}} = \max(5 \times 10^{-7}, 100A(0)). \quad (23)$$

The transition is smoothed in logarithmic amplitude,

$$w(t) = s_0 \left[1 + \exp \left(-10 \log \frac{A(t)}{A_{\text{tr}}} \right) \right]^{-1}. \quad (24)$$

We set $w(t) = 0$ when $A(t) = 0$. Thus $w = 1/2$ at the threshold, while $w \approx 10^{-10}$ when $A/A_{\text{tr}} = 0.1$. This avoids a discontinuous switch in the numerical flow.

Once the envelope grows, both canonical variables are multiplied after each Gauss–Legendre substep by a blended eighth-order Hou–Li-type filter [13],

$$\sigma_h(k) = \exp \left[-0.69 \left(\frac{|k|}{k_{\text{eff}}(h)} \right)^8 \right], \quad k_{\text{eff}}(h) = \min \left(128, \frac{12}{h} \right), \quad (25)$$

$$\hat{u}_k^+ = [(1 - w) + w\sigma_h(k)] \hat{u}_k^-, \quad u \in \{\eta, \xi\}. \quad (26)$$

The scale $k_{\text{eff}}h \leq 12$ follows the depth scaling of a solitary wave’s horizontal width. It is conservative relative to the truth-spectrum audit: across the 64 Tanaka reference rollouts, removing content with $|k|h > 8$ changes η and ξ by at most 1.22×10^{-9} and 2.94×10^{-11} , respectively, at the 99th percentile. For the failing depth $h = 0.17760$, $k_{\text{eff}} = 67.57$; for the shallow phase-error case, $12/h > 128$, so no additional depth restriction is introduced. The guard is therefore a conditional spectral viscosity, not a hard global cutoff.

3 Historical v9 data used for the reported checkpoints

3.1 Snapshot construction

All checkpoint results reported in this manuscript use the historical `combined_dataset_v9.npz` archive. The parameterized replacement dataset discussed below has not yet been used to train a reported model. Each historical training example is a tuple

$$(\eta_j, \xi_j, h_j) \mapsto q_j^*, \quad (27)$$

on the $N = 1024$ grid, where q_j^* is the stored reference target under that source’s generation protocol. Initial conditions are sampled by wave family and integrated with the double-precision reference method of [section 2](#). For rollout-derived sources we save multiple trajectory times, recompute the order-six, pad-eight DNO label in manageable chunks, and project the label to $|k| \leq 128$. Three static sources (IDs 2–4) instead store their unfiltered order-six labels. The mode loss uses only $1 \leq k \leq 128$, where the conventions agree; the base loss also supervises the stored high modes of the static sources. Fields are stored as 32-bit arrays, while trajectory generation and label evaluation use float64.

The current v9 dataset contains 7 633 989 snapshots and occupies approximately 88 GB. Its composition is shown in [table 1](#). The families deliberately combine coherent solutions, spectrally broad random fields, shallow and finite-depth cases, and nonlinear modulation, broadening the training support beyond solitary-wave states.

Table 1: Construction-category composition of the historical `combined_dataset_v9.npz`, with independent generator shards pooled.

Construction category	Snapshots	Construction category	Snapshots
Random sea	460,740	Manufactured harmonic	500,000
Stokes	1,000,000	Tanaka	2,200,000
BF-inspired wavetrain	2,033,249	Sparse Fourier packets	1,440,000
Total			7,633,989

The archive uses fourteen source identifiers to record the generating shard. In particular, suffixes such as `g0`, `g1`, and `modal` identify independent generation shards or entry points. They do not denote different DNO truncation orders, target definitions, or physical wave families. We pool those shards in [table 1](#).

3.2 Three roles of the historical construction categories

The six historical construction categories in [table 1](#) are grouped below according to the states they contribute. This grouping is independent of the GPU or generator shard that produced a snapshot.

Coherent traveling and periodic waves (3,200,000 snapshots). This group contains both Tanaka and Stokes sources. Single- and multi-crest Tanaka conditions are constructed from numerical solitary-wave profiles, with crest amplitude sampled relative to depth and positions randomized subject to a minimum separation. The main shallow set reaches $h = 0.01$ and uses an occupancy-aware upper bound to avoid cutting a broad tail at the periodic boundary. The steep addition contributes 200,000 snapshots in $h \in [0.20, 0.35]$ and $a/h \in [0.25, 0.45]$. Deep- and finite-depth Stokes waves supply periodic states with controlled harmonic content. Together these sources resolve coherent nonlinear structure across depth without treating the deep/finite data partitions as separate physical families.

JCP09-inspired modulated wavetrains (2,033,249 snapshots). The historical constructor starts from a fifth-order Stokes carrier and two sidebands inspired by [\[16\]](#), but it is not the literal JCP09 initial condition. It uses one carrier coefficient for both sidebands, adds empirical cross terms, and applies the deep-water sideband formula at sampled finite depths. Only 46.7% of the archived parameter draws lie inside the leading deep-water modulational-instability band. The three archived shards are independent samples from this same construction; the suffix `modal` does not define a separate family.

Spectral-coverage and perturbation states (2,400,740 snapshots). This group combines the manufactured-harmonic, random-sea, and three sparse Fourier-packet sources. Manufactured harmonics anchor the small-amplitude limit, while random seas broaden phase, bandwidth, and local-slope coverage. The sparse packet sources address the neighborhood of solitary waves rather than the exact solitary-wave manifold: they combine one to five Fourier modes while controlling kh , a/h , and ka .

3.3 Forward paper-dataset design

The definitions in this subsection belong to a replacement dataset that has not yet been used for training. In particular, they do not describe a retrospective cleaning of v9 and cannot explain the errors of the checkpoints reported in this manuscript. The replacement has four physical families: fifth-order Stokes states, periodic tangent-Hermite Tanaka profiles, the equation-(33) sideband construction paired with the project’s analytic fifth-order deep-water carrier, and resolved-band JONSWAP/TMA random seas. The four physical constructors and their parameter samplers are implemented. The current family revisions are 2 for Stokes, 3 for Tanaka, and 4 for both Benjamin–Feir and JONSWAP/TMA. A revision number is therefore local to a physical family, rather than a single dataset-wide version number. Unless an endpoint convention is written explicitly, closed brackets in a displayed parameter box denote the closure of the continuous support. The ordinary floating-point uniform draw uses its lower endpoint and excludes its upper endpoint. An endpoint convention written explicitly in an individual sampling formula—including the half-open phase and center intervals and the mixed open–closed Benjamin–Feir steepness draw—overrides that default. These endpoint choices do not change a continuous uniform law, but they are specified for deterministic regeneration. For each trajectory family, one common executor runs a residual-controlled GL2 trajectory with time step 0.01, saving a candidate state every 0.08. JONSWAP/TMA first uses the explicitly time-dependent adjustment defined below and then starts its autonomous production trajectory. A case is retained only when its declared numerical path returns a complete admissible trajectory on the requested interval. The 0.005 and 0.0025 steps are used only by a separate method-validation utility; they are not additional production rollouts or retry steps.

Common modules assign a split and parameter category before numerical work and write one atomic completed-batch artifact containing the family and split, the ordered parameter-group identifiers, and the accepted row arrays. A batch slot with no owned row block denotes a rejected attempt; sampled specifications and phases, failure reasons, and diagnostic metrics are not persisted. A resumable driver continues until the accepted-case quota in every parameter category is met, replacing a rejected attempt only in its original category. The fixed split, family, and attempted-simulation index seed deterministic regeneration. The training loader uses the preassigned splits and globally shuffles all accepted training rows. Unexpected construction or target-evaluation errors propagate rather than being silently converted into ordinary rejected samples. The family laws below therefore describe attempted specifications, not the unconditional law of loader-visible cases. If X is a complete attempted specification in preassigned category i , A_f is the declared complete case-acceptance event for family f , and w_i is the accepted-quota weight, then the released law is

$$p_{\text{release}}(i, x) = w_i p_{\text{proposal}}(x \mid i, A_f = 1). \quad (28)$$

For Stokes, A_f comprises declared static support, a finite state, positive water column, and a finite target. For Tanaka, it comprises declared construction support followed by complete autonomous-trajectory acceptance; for Benjamin–Feir, initial-state validity followed by complete autonomous-trajectory acceptance. For JONSWAP/TMA, it comprises initial construction and graph validity followed by nonlinear adjustment and complete autonomous- trajectory acceptance. Thus the quotas preserve the declared category marginals, not the unconditioned proposal law within a category. This is a statement about how the release is constructed; it does not by itself assert a measured within-category shift for every family. No retrospective parameter cutoff is applied. The loader-facing manifest and row map record the selected shards, split totals, and case-to-row ownership; they do not persist numerical- configuration or source hashes. The release described here selects Stokes revision 2, Tanaka revision 3, Benjamin–Feir revision 4, and JONSWAP/TMA revision 4.

An earlier generator-revision-1 complete transaction smoke test used a reduced wiring contract: 64 spatial grid points, Craig–Sulem order zero, padding factor one, largest delivered wavenumber 16, and the former paired 0.04/0.02 validation path. One Tanaka case, one Benjamin–Feir case, and one random-sea case were each accepted to a quota of one case; each committed three rows, and all nine rows were finite when read by the training loader. This reduced calculation tests the transaction and loader, not the current family-revision executors, the paper target, the long-horizon numerical contract, or a population acceptance rate. The current numerical paths have focused software-test coverage. A former revision-2 104-category trajectory pilot predates the Tanaka revision-3 support and numerical contract and remains historical software-path evidence. The present family specifications contain 108 categories: four Stokes, eleven Tanaka, 66 Benjamin–Feir, and 27 JONSWAP/TMA. The exact revision-3 Tanaka pilot passed all eleven Tanaka categories. The revision-4 Benjamin–Feir exact-source gate passed all 66 categories, and its production splits are complete at 16384 training cases and 1024 cases in each of validation and test. The revision-4 JONSWAP/TMA source was tested more broadly: a predeclared gate retained 540 trajectories, 20 in each of its 27 categories, from 548 attempts. Its observed rejection rate was therefore $8/548 = 1.46\%$, below the predeclared two-percent bound. This was the completed constructor-adoption criterion, not a bulk release threshold. The subsequent revision-4 bulk release is complete: it retained 18 432 cases from 18 726 attempts, recorded 294 zero-row rejections, and stored 294 912 rows.

The revision-2 Stokes splits are complete at the same accepted-case counts as Benjamin–Feir. A historical Tanaka run produced 2048 training, 1024 validation, and 1024 test cases, but those trajectories are excluded from the paper dataset. The old solitary-wave solve could not realize crest amplitudes below 9.9478×10^{-4} ; 64 of the 4096 historical cases contain one requested component below this floor. The smallest requested component was inflated by a factor 41.27 after periodic construction and projection. The corrected solve uses $q_c^{\max} = 1 - 10^{-12}$, 48 bisections, and checks that every realized crest height matches its request. The corrected revision-3 release is a fresh deterministic population with 16384 training cases and 1024 cases in each of validation and test. It does not reuse the historical random specifications, and no old and corrected Tanaka rows are mixed. All 18 432 corrected attempts were accepted and stored 3 686 400 rows.

One frozen learning target. Let the spatial period be $L = 2\pi$, let $N = 1024$, and let $\Delta x = L/N$ and $x_\ell = \ell\Delta x$, $\ell = 0, \dots, N - 1$, be the grid spacing and nodes. For a periodic grid function u , let $P_K u$ retain its Fourier modes with $|k| \leq K$, where $K = 128$, and let $P_K^\circ u$ also set the zero Fourier mode to zero. We write $\mathcal{G}_M^{N,p}(\eta; h)\xi$ for the order- M Craig–Sulem approximation on N points with product-padding factor p , surface elevation η , surface potential ξ , and depth $h > 0$. The target is fixed to

$$q_{\text{ref}}(\eta, \xi; h) = P_K^\circ \left[\mathcal{G}_6^{1024,8}(P_K \eta; h)(P_K \xi) \right]. \quad (29)$$

Thus both inputs are restricted before evaluation, and the output is restricted and made mean zero afterward. The target is evaluated in float64; the three fields η , ξ , and q_{ref} are stored in float32, while depth and retained time are stored in float64. The sampled parameters from which a row was constructed are not stored. Thus the mean-zero statement is exact for the float64 target and is preserved in the stored array only to float32 rounding error. Static Stokes states use exactly [equation \(29\)](#), rather than the unfiltered labels in some historical v9 sources. The paper generator fixes L, N, K, M, p and the evaluation precision in its numerical settings; the completed batches do not record target-configuration or source hashes. The three trajectory families have internal evolution contracts specified below, but all deliver the same target in [equation \(29\)](#).

JONSWAP/TMA is evolved on a larger grid than the stored target. To specify the transfer

without an FFT-normalization ambiguity, for a field sampled on N' points define

$$\hat{f}_k^{(N')} = \frac{1}{N'} \sum_{j=0}^{N'-1} f\left(\frac{2\pi j}{N'}\right) e^{-2\pi i j k / N'}. \quad (30)$$

At each retained JONSWAP/TMA time, the coefficients of η and ξ with $|k| \leq 128$ are copied from the 2048-point internal grid and reconstructed on the 1024-point grid. Equation (29) is then evaluated from those reconstructed fields. The internal value of $G(\eta; h)\xi$ is never transferred between grids; recomputing it is necessary because the stored label is the declared 1024-point order-six operator, not the order-four operator used for evolution.

Static fifth-order Stokes states. Let $n \geq 1$ be an integer carrier mode, $k = 2\pi n/L$ its wavenumber, $a > 0$ the amplitude parameter in Fenton's fifth-order expansion [10], $h > 0$ the depth, and $\phi \in [0, 2\pi)$ a spatial phase. The finite- and deep-water coefficient tables are two parameter branches of the same fifth-order family. In either branch the constructor evaluates the state directly at

$$\theta = kx + \phi, \quad \phi \sim \text{Unif}[0, 2\pi). \quad (31)$$

No sampled time is used as a substitute for ϕ : changing a changes the nonlinear temporal frequency, whereas equation (31) preserves the requested spatial translation. The spatial mean of ξ is removed to fix the irrelevant constant-potential gauge.

The common amplitude interval is

$$0.000766 \leq a \leq 0.011494. \quad (32)$$

In the finite-depth branch,

$$n \in \{14, \dots, 26\}, \quad 0.02 \leq h \leq 1.5, \quad 0.5 \leq kh \leq 5, \quad (33)$$

whereas the deep-water branch uses

$$n \in \{1, \dots, 20\}, \quad 4 \leq h \leq 50, \quad kh \geq 5. \quad (34)$$

The steepness ka is assigned to one of the two categories

$$0.005 \leq ka < 0.03, \quad 0.03 \leq ka \leq 0.15. \quad (35)$$

The finite/deep branches crossed with these two intervals form four parameter categories. Within an assigned category, the sampler first enumerates the carrier modes for which the branch, depth, amplitude, and steepness intervals have nonempty intersection, and chooses one of those modes uniformly. It then draws h log-uniformly on the mode-conditioned depth interval, draws ϕ uniformly on $[0, 2\pi)$, and draws a uniformly from the intersection of equation (32) with the assigned steepness interval.

The finite-depth branch has an additional support condition. Write its elevation as

$$\eta_5(\theta) = \sum_{m=1}^5 E_m \cos(m\theta), \quad (36)$$

where E_m is the coefficient of the m th harmonic. Let $H = \max_{\theta} \eta_5(\theta) - \min_{\theta} \eta_5(\theta)$ be the exact crest-to-trough height, and let H_{4096} be the corresponding maximum minus minimum over 4096 equally spaced values of θ . Define

$$H_+ = H_{4096} + \frac{1}{4} \left(\frac{2\pi}{4096} \right)^2 \sum_{m=1}^5 m^2 |E_m|, \quad \text{Ur}_+ = \frac{H_+ \lambda^2}{h^3}, \quad \lambda = \frac{2\pi}{k}. \quad (37)$$

Since $|\eta_5''| \leq \sum_{m=1}^5 m^2 |E_m|$, Taylor’s theorem at the true maximum and minimum gives $H \leq H_+$. The finite-depth sampler therefore requires the conservative condition $\text{Ur}_+ \leq 26$. This is a restriction on the parameter support of the fifth-order family, not a filter trained on integration failures. Fenton identifies the corresponding shallow-water expansion parameter for fifth-order Stokes theory [10], and Zhao, Wang, and Liu use $\text{Ur} = 26$ to separate the fifth-order Stokes and shallower cnoidal regimes [20]. The coefficient implementation follows the fifth-order formula checked against Zhao and Liu [19]. The deep-water branch instead requires $kh \geq 5$.

Seeded sampling. Let s be the root seed fixed by the data split, let f identify the physical family, and let i be the attempted-simulation index. A simulation uses NumPy PCG64 initialized from (s, f, i) . The four categories are the finite/deep branches crossed with the two intervals in equation (35). Requested accepted-case counts are divided among these categories by quotient and remainder before any state is constructed. This is a seeded stratified pseudorandom construction, not a low-discrepancy sequence. A successful proposal holds its realized mode, amplitude, depth, phase, branch, and category only long enough to construct its rows; these values can instead be regenerated from the split, family, and attempt seed.

If a finite-depth amplitude draw violates $\text{Ur}_+ \leq 26$, only the amplitude is redrawn, uniformly from the same category intersection. Its mode, depth, phase, and category remain fixed. The case may make at most 1000 additional amplitude draws. Intermediate rejected amplitudes and their values of Ur_+ are not persisted. If the redraw allowance is exhausted, the sampler returns no state and the batch slot owns no rows; generation then fills the remaining accepted-case quota from a later attempt in the same category.

Static archive and replay checks. These checks concern only the implemented Stokes component of the replacement dataset. A 16-case finite-depth archive contained eight cases from each category in equation (35), required three same-category redraws, and had $\max \text{Ur}_+ = 18.0578589$. A separate 16-case deep archive was finite and had $\min kh = 10.0678$. Its category counts were nine and seven because some carrier modes admitted only one steepness category; the subset admitting both remained balanced to within one. Re-evaluating equation (29) from the stored float64 parameters and then converting the fields to float32 reproduced both archives bit for bit. A synthetic interruption after two batches likewise resumed to the same 30 ordered archive entries, byte for byte, as an uninterrupted 32-case run. In a 20,000-case finite-depth sampling stress with root seed 20260725 and stream zero, the two category counts were 10,000 and 10,000, the largest accepted value was $\text{Ur}_+ = 25.9724665$, the largest redraw count was 56, and no case exhausted the cap. A separate forced-exhaustion quota test committed the attempt as a zero-row out-of-support rejection and scheduled a replacement in the same category.

The common static transaction was also exercised at the exact target resolution $(N, M, p, K) = (1024, 6, 8, 128)$ in float64. One validation attempt was assigned to each finite/deep and low/moderate category, and its complete specification was written before construction. All four attempts had finite input and target fields, positive water column, no support violation, and no failed required quality check. The transaction therefore committed one $t = 0$ row per case and constructed the split-aware dataset view. This accepted-quota calculation accepted all four attempts and checks the exact implementation path; it does not estimate a population acceptance probability.

Periodic tangent-Hermite Tanaka states. Let $\alpha = a/h > 0$ be the ratio of crest amplitude a to depth h . Following Tanaka’s solitary-wave formulation [15], a unit-depth solve at amplitude ratio α returns a dimensionless surface profile $\bar{\eta}_\alpha(X)$, a tabulated surface angle $\theta_\alpha(X)$, and a Froude

number F_α . Xu and Guyenne also use a solitary wave computed by Tanaka's method [16]. The tangent-Hermite interpolation, periodization, multi-crest superposition, and parameter sampling below are choices of the present dataset; they are not parts of either cited solitary-wave construction. Here X is the dimensionless horizontal coordinate, and

$$\frac{d\bar{\eta}_\alpha}{dX}(X) = \tan \theta_\alpha(X). \quad (38)$$

Before interpolation, the constructor verifies that both endpoint elevations and the endpoint and crest tangents differ from zero by at most 10^{-12} , then sets those quantities exactly to zero. The corrected tabulated profile is extended by zero outside its finite tabulation interval. Between adjacent tabulation points, the constructor uses the unique cubic polynomial matching both endpoint values of $\bar{\eta}_\alpha$ and both endpoint values of $\tan \theta_\alpha$. Thus “tangent-Hermite” specifies the interpolation, not an additional wave model. The physical derivative is correctly scaled because

$$\partial_x \left[h \bar{\eta}_\alpha \left(\frac{x - x_0 + rL}{h} \right) \right] = \frac{d\bar{\eta}_\alpha}{dX} \left(\frac{x - x_0 + rL}{h} \right). \quad (39)$$

For a crest centered at $x_0 \in [0, L)$, define the periodic elevation

$$\eta_{\alpha,h,x_0}(x) = h \sum_{r \in \mathbb{Z}} \bar{\eta}_\alpha \left(\frac{x - x_0 + rL}{h} \right). \quad (40)$$

Only finitely many terms in equation (40) are nonzero at a given x , because the extended profile has finite support. The implementation sums every periodic image whose support intersects $[0, L)$; it does not retain only the nearest three images. The sum is evaluated on an $8N$ -point grid and Fourier-restricted to the N -point grid.

Let $g > 0$ be gravitational acceleration, let $c_\alpha = F_\alpha \sqrt{gh}$ be the unsigned wave speed, and let $s \in \{-1, 1\}$ denote the direction of travel. Write $\eta_x = \partial_x \eta_{\alpha,h,x_0}$. The traveling-wave relation first defines

$$r_{\alpha,h,x_0,s}(x) = s \left[c_\alpha - \sqrt{(1 + \eta_x(x)^2)(c_\alpha^2 - 2g\eta_{\alpha,h,x_0}(x))} \right]. \quad (41)$$

Define the expression under the square root by

$$\mathcal{R}_{\alpha,h,x_0}(x) = (1 + \eta_x(x)^2)(c_\alpha^2 - 2g\eta_{\alpha,h,x_0}(x)). \quad (42)$$

The constructor requires the nodewise condition $\min_\ell \mathcal{R}_{\alpha,h,x_0}(x_\ell) \geq 0$ and treats an attempted component that violates this condition as inadmissible; it does not replace a negative value by zero. For the fixed upper-steepness, seam-centered test case, the initial value is $\min_\ell \mathcal{R}_{\alpha,h,x_0}(x_\ell)/c_\alpha^2 = 0.372151$. Hence the clipping in the old writer did not change that particular initial condition. This single positive margin does not establish the condition for the full population.

Let $\hat{r}_n$ be the Fourier coefficient of $r_{\alpha,h,x_0,s}$ at wavenumber $k_n = 2\pi n/L$. The periodic, zero-mean surface potential is defined by

$$\hat{\xi}_n = \begin{cases} \hat{r}_n/(ik_n), & n \neq 0, \\ 0, & n = 0. \end{cases} \quad (43)$$

Consequently $\partial_x \xi = r_{\alpha,h,x_0,s} - \hat{r}_0$; subtracting the constant $\hat{r}_0$ is necessary for a periodic antiderivative.

A case with $m \geq 1$ crests has parameters $(h, m, \alpha_j, x_j, s_j)_{j=1}^m$ and initial state

$$(\eta_0, \xi_0) = P_K \sum_{j=1}^m (\eta_{\alpha_j,h,x_j}, \xi_{\alpha_j,h,x_j,s_j}). \quad (44)$$

For $m > 1$, [equation \(44\)](#) is a superposition of separately constructed solitary-wave profiles, not an exact steady multi-solitary-wave solution. The sharp $P_K = P_{128}$ map in [equation \(44\)](#) is applied once before time integration. Revision 3 then embeds this state in the larger $|k| \leq 256$ evolution space by setting modes $129 \leq |k| \leq 256$ initially to zero. Nonlinear evolution may populate those workspace modes, but the initial-value problem is exactly the sharp- P_{128} problem used in the resolution calculation below. For two centers $x_i, x_j \in [0, L]$, define their periodic distance by

$$d_L(x_i, x_j) = \min_{r \in \mathbb{Z}} |x_i - x_j + rL|. \quad (45)$$

The centers must satisfy $d_L(x_i, x_j) \geq 3h$ whenever $i \neq j$.

The amplitudes are sampled before the depth. In a main group, define the total dimensionless amplitude $\alpha_\Sigma = \sum_{j=1}^m \alpha_j$ and draw $\alpha_\Sigma \sim \text{Unif}[0.10, 0.35]$. Draw nonnegative weights $(\lambda_1, \dots, \lambda_m)$ uniformly on the simplex $\sum_j \lambda_j = 1$, equivalently from $\text{Dirichlet}(1, \dots, 1)$, and set $\alpha_j = \alpha_\Sigma \lambda_j$. For $m = 1$, this means $\lambda_1 = 1$. In a steep group, draw the single amplitude ratio $\alpha_1 \sim \text{Unif}[0.25, 0.45]$. In both groups, define

$$\alpha_{\max} = \max_{1 \leq j \leq m} \alpha_j. \quad (46)$$

The depth interval is then restricted before a profile is constructed. The small-amplitude KdV profile has the form

$$\frac{\eta(x)}{h} \approx \alpha_{\max} \text{sech}^2(\kappa(x - x_0)), \quad \kappa = \frac{\sqrt{3\alpha_{\max}}}{2h}, \quad (47)$$

where $\text{sech } z = 1/\cosh z$ and κ is an inverse length. We use κ only as a small-amplitude KdV inverse-width proxy for the narrowest component. It is not the exact width of a Tanaka profile. With the delivered cutoff $K = 128$, define the number of delivered wavenumbers per proxy inverse width by

$$R = \frac{128}{\kappa} = \frac{256h}{\sqrt{3\alpha_{\max}}}. \quad (48)$$

Tanaka revision 3 requires $R \geq 10$. This threshold is a numerical design choice established by the resolution tests below; it is not a condition stated in [\[15\]](#) or [\[16\]](#).

The proxy first arose from a retrospective resolution audit, rather than from a fitted rejection statistic. We reconstructed 256 tangent-Hermite proposals before their sharp P_{128} projection. All eleven cases that exceeded the historical $K_{\text{evol}} = 224$ Hamiltonian-drift diagnostic had $R < 8$. The 210 cases with $R \geq 8$ contained none of those diagnostic failures, and their largest raw-to- P_{128} relative root-mean-square errors were 6.80×10^{-5} for η and 4.47×10^{-6} for ξ . By comparison, the unrelated fixed restriction $h \geq 0.025$ admitted a larger maximum η projection error, 4.33×10^{-4} . These observations identify under-resolved input width as the mechanism; the full-horizon panel below, not the historical Hamiltonian threshold, determines the revision-3 support.

For a base lower depth $b > 0$, define

$$h_-(\alpha; b) = \max \left\{ b, \frac{10\sqrt{3\alpha}}{256} \right\}. \quad (49)$$

After the amplitudes have been drawn, sample $U_h \sim \text{Unif}[0, 1]$ and set

$$h = h_-^{1-U_h} h_+^{U_h}, \quad (50)$$

where $h_- = h_-(\alpha_{\max}; 0.01)$ and $h_+ = 0.30$ in a main group, while $h_- = h_-(\alpha_1; 0.20)$ and $h_+ = 0.35$ in a steep group. Thus h is log-uniform on the conditional interval $[h_-, h_+]$. The resolution term in

equation (49) is at most 0.0454 on the steep amplitude interval, so the steep lower bound remains 0.20 and the resolution restriction is inactive there. The resulting supports are therefore

$$\begin{aligned} m \in \{1, 2, 3\}, \quad \alpha_\Sigma \in [0.10, 0.35], \quad h \in [h_-(\alpha_{\max}; 0.01), 0.30] & \quad (\text{main}), \\ m = 1, \quad \alpha_1 \in [0.25, 0.45], \quad h \in [0.20, 0.35] & \quad (\text{steep}). \end{aligned} \quad (51)$$

Equal increments of U_h give equal multiplicative changes of depth. This is a coverage rule for the numerical experiment, not a probability model for physical water depths. In a deterministic million-draw audit for each $m = 1, 2, 3$, the $R \geq 10$ condition removed 28.55% of the former independent-depth law after weighting the nine main categories by their 2 : 3 : 4 category counts. The median depths for $m = 1, 2, 3$ were respectively 0.0973, 0.0902, and 0.0857, and their 95th percentiles were 0.2680, 0.2659, and 0.2647.

For each case let

$$q = \#\{j : s_j = +1\} \quad (53)$$

be the number of right-moving crests. The three choices of m in equation (51), crossed with $q = 0, \dots, m$, give nine parameter categories. The steep support in equation (52), crossed with $q = 0, 1$, gives two more. Thus the Tanaka family has eleven parameter categories.

The requested Tanaka accepted-case count is divided equally across these eleven categories, up to the unavoidable difference of one case when the total is not divisible by eleven. Conditional on q , the sampler chooses uniformly which q of the m exchangeable crest labels have sign +1; all remaining signs are -1 . Hence left and right directions remain balanced marginally in the ideal equal-category population, while the direction compositions $q = 0, \dots, m$ receive equal coverage instead of their former binomial frequencies under independent fair signs. For a finite quota, the quotient-remainder assignment changes category counts by at most one and may therefore introduce a one-case direction imbalance. In particular, the fixed category order gives a quota of 1024 one extra left-moving one-crest case. This law also gives more aggregate Tanaka weight to $m = 2$ and $m = 3$, because they have three and four distinct direction compositions, respectively. The corrected full-size Tanaka production is complete. Its historical 4096-case predecessor and its eleven-case revision-3 pilot are excluded from the replacement-dataset view.

Tanaka revision 3 retains the eleven direction-aware categories introduced in revision 2, but changes the amplitude–depth draw order, the admissible depth support, and the internal numerical contract. Earlier Tanaka revision-1 and revision-2 smokes, dry runs, and pilot artifacts remain historical constructor or transaction evidence; they cannot be resumed into a revision-3 run. Benjamin–Feir and JONSWAP/TMA use revision 4; static Stokes remains at revision 2.

To sample centers without a translation bias, let $(w_1, \dots, w_m)$ have the same uniform-simplex law and set the cyclic gaps to

$$g_j = 3h + (L - 3mh)w_j, \quad j = 1, \dots, m. \quad (54)$$

A uniform global rotation places the resulting point set on the periodic domain. Directions are then assigned according to the category as described above. In a steep group, the center uses this same uniform rotation and the direction is fixed by $q \in \{0, 1\}$. The separation is always feasible on the declared support because $3mh \leq 2.7 < 2\pi = L$. Thus separation and translation symmetry are enforced before integration.

The population sampler and ordered batch adapter are implemented. The sampled parameters are held only during construction and can be regenerated from the split, family, and attempt seed. The potential routine transfers the computed radicand to the host once before taking the square root and introduces no clipping tolerance. An everywhere-finite, positive-speed component with

a negative radicand is a declared support rejection: it contributes no rows and is replaced in the same category. A nonfinite radicand, a nonpositive or nonfinite computed speed, or an unexpected exception remains an implementation error and stops the run. In the historical reduced paired calculation described above, the Tanaka attempt was accepted, committed three finite rows, and was finite when read by the training loader. This is transaction evidence at the reduced contract, not evidence about the current Tanaka trajectory executor or the Tanaka population.

Tanaka revision 3 evolves an internal band $|k| \leq K_{\text{evol}}$ with $K_{\text{evol}} = 256$. Let $\hat{\eta}_{0,k}$ and $\hat{\xi}_{0,k}$ be the Fourier coefficients of the sharp- P_{128} state in [equation \(44\)](#). The internal state is initialized by the zero-filled lift

$$(\hat{\eta}_k(0), \hat{\xi}_k(0)) = \begin{cases} (\hat{\eta}_{0,k}, \hat{\xi}_{0,k}), & |k| \leq 128, \\ (0, 0), & 128 < |k| \leq 256. \end{cases} \quad (55)$$

Thus no discarded raw-profile mode is supplied to the wider evolution at $t = 0$; modes above 128 can arise only from the subsequent dynamics. The $K_{\text{evol}} = 224$ comparison arm uses the same construction with upper limit 224. After every completed GL2 step, the Fourier coefficients $\hat{\eta}_k$ and $\hat{\xi}_k$ are multiplied by

$$\sigma_k = \exp \left[-36 \left(\frac{|k|}{256} \right)^{36} \right]. \quad (56)$$

This is the power-36 Hou–Li filter [13] used on both canonical variables, with the parameters of [16, Eq. (29)]. The $K_{\text{evol}} = 224$ comparison arm replaces 256 in [equation \(56\)](#) by 224. At each saved time, the delivered state is $(P_{128}\eta, P_{128}\xi)$ and the delivered label is q_{ref} from [equation \(29\)](#). Modes $129 \leq |k| \leq 256$ therefore stabilize the internal evolution but are not stored as inputs or labels for learning. This sharp P_{128} initial projection, zero-filled lift to $K_{\text{evol}} = 256$, post-step Hou–Li multiplier, and final P_{128} delivery is the exact revision-3 contract used by the pilot. The excluded historical July training shards used the same numerical contract but are not part of the fresh paper dataset.

The resolution choice was tested on nine one-crest cases. Their amplitude ratios were $\alpha \in \{0.225, 0.30, 0.35\}$, and their resolution ratios were $R \in \{6, 8, 10\}$. Each case was evolved from the same bit-identical P_{128} initial state with $K_{\text{evol}} = 224$ and 256 through $T = 200$. Define the dimensionless delivered fields

$$\tilde{\eta}_J(t) = \frac{P_{128}\eta_J(t)}{h}, \quad \tilde{\xi}_J(t) = \frac{P_{128}\xi_J(t)}{h\sqrt{gh}}, \quad \tilde{q}_J(t) = \frac{q_{\text{ref}}(\eta_J(t), \xi_J(t); h)}{\sqrt{gh}}, \quad (57)$$

where $J \in \{224, 256\}$ is the internal cutoff and (η_J, ξ_J) is that arm’s internal state. For $u \in \{\tilde{\eta}, \tilde{\xi}, \tilde{q}\}$, define the reported cross-resolution error by

$$\epsilon_u = \max_t \frac{\|u_{224}(t) - u_{256}(t)\|_{2,N}}{\max_s \|u_{256}(s)\|_{2,N} + 10^{-12}}, \quad \|v\|_{2,N}^2 = \frac{1}{N} \sum_{\ell=0}^{N-1} |v(x_\ell)|^2. \quad (58)$$

The three fields are nondimensionalized before the additive 10^{-12} floor is applied. Write ϵ_η , ϵ_ξ , and ϵ_q for [equation \(58\)](#) evaluated with $u = \tilde{\eta}$, $\tilde{\xi}$, and $\tilde{q}$, respectively. The audit also checked the analogous full-history RMS error. To measure whether hidden state modes alter the delivered operator, define

$$d_J(t) = P_{128}^\circ \left[\mathcal{G}_6^{1024,8}(\eta_J(t); h) \xi_J(t) \right] - q_{\text{ref}}(\eta_J(t), \xi_J(t); h), \quad (59)$$

where (η_J, ξ_J) is the internal state of arm J . Its reported closure error is

$$C_J = \max_t \frac{\|d_J(t)\|_{2,N}}{\max_s \|q_{\text{ref}}(\eta_J(s), \xi_J(s); h)\|_{2,N} + 10^{-12}}. \quad (60)$$

All three $R = 6$ controls failed the predeclared 10^{-3} comparison; their worst operator error was $\epsilon_q = 9.36 \times 10^{-3}$ and their worst closure error was 2.91×10^{-3} . All three $R = 8$ cases passed, but their worst errors were $(\epsilon_\eta, \epsilon_\xi, \epsilon_q) = (1.87 \times 10^{-4}, 1.29 \times 10^{-5}, 7.34 \times 10^{-4})$ and $\max_J C_J = 6.31 \times 10^{-4}$. At $R = 10$, these values fell to $(2.94 \times 10^{-5}, 1.62 \times 10^{-6}, 1.35 \times 10^{-4})$ and 1.40×10^{-4} . Each arm completed all 20,000 GL2 time steps, and both coupled implicit stage equations converged at every step. The largest delivered- P_{128} Hamiltonian drift among the $R = 10$ arms was 2.99×10^{-8} ; this value is reported only as a numerical diagnostic and is not a production acceptance condition. We therefore use the round margin $R \geq 10$ and the exact two-times internal band $K_{\text{evol}} = 256$ in Tanaka revision 3.

Equation-(33) Benjamin–Feir states. This family is restricted to deep water. Let $n_c \in \mathbb{N}$ be the carrier-mode index, let $\Delta n \in \mathbb{N}$ be the symmetric sideband offset with $\Delta n < n_c$, let $\varepsilon_c > 0$ be the steepness of the first elevation harmonic, let $\rho > 0$ be the sideband-to-carrier amplitude ratio, and let $x_0 \in [0, L)$ be a global spatial translation. Thus ε_c is not defined from half the crest-to-trough wave height. Define

$$k_c = \frac{2\pi n_c}{L}, \quad k_- = \frac{2\pi(n_c - \Delta n)}{L}, \quad k_+ = \frac{2\pi(n_c + \Delta n)}{L}, \quad a = \frac{\varepsilon_c}{k_c}. \quad (61)$$

The sideband ansatz follows equation (33) of Xu and Guyenne, who in turn follow Dommermuth and Yue [6, 16]. Their published test uses, in the present notation,

$$(n_c, \Delta n, \varepsilon_c, \rho, \theta) = (9, 2, 0.13, 0.10, -\pi/4), \quad L = 2\pi, \quad g = 1. \quad (62)$$

It also uses a numerically computed exact steady Stokes carrier. The construction below is our randomized generalization over 66 integer mode-pair categories, not a reproduction of that single parameter tuple.

Let $y = x - x_0$, and let $(\eta_c(y), \xi_c(y))$ denote the project’s analytic fifth-order deep-water Stokes formula, with carrier phase zero in this translated coordinate and first elevation harmonic of amplitude a . This analytic fifth-order carrier is an approximation to the published exact steady carrier; it is not the Xu–Guyenne carrier itself. Both sidebands use the fixed relative phase $\theta = -\pi/4$ from the cited calculation.

With η_0 evaluated before ξ_0 , the implemented sideband construction is

$$\begin{aligned} \eta_0(x) &= \eta_c(y) + \rho a [\cos(k_- y + \theta) + \cos(k_+ y + \theta)], \\ \xi_0(x) &= \xi_c(y) + \rho a \left[\sqrt{\frac{g}{k_-}} e^{k_- \eta_0(x)} \sin(k_- y + \theta) + \sqrt{\frac{g}{k_+}} e^{k_+ \eta_0(x)} \sin(k_+ y + \theta) \right]. \end{aligned} \quad (63)$$

The spatial mean of ξ_0 is removed after equation (63). Both sidebands use the same ρ and θ ; there are no empirical cross terms. Translating the complete state changes neither the relative phases nor the resonant quartet phase, which remains $\theta + \theta - 2(0) = -\pi/2$. The numerical depth is $h = 5L/(2\pi)$, so the first nonzero wavenumber $k_1 = 2\pi/L$ satisfies $k_1 h = 5$.

For a proposed pair, define the normalized sideband location and a focused-envelope steepness proxy by

$$\beta = \frac{\Delta n/n_c}{2\sqrt{2}\varepsilon_c}, \quad \varepsilon_f = \varepsilon_c \left(1 + 2\sqrt{1 - \beta^2} \right). \quad (64)$$

Benjamin and Feir give the historical deep-water instability condition [2]. In the present normalization, equation (4.6) of Andrade and Stuhlmeier gives $0 < \beta < 1$, while their equations (5.7)–(5.11) give the Akhmediev-breather correspondence and the peak-envelope amplification factor $1 + 2\sqrt{1 - \beta^2}$ [1].

We use that factor only to define the scalar population-screening proxy above. Yang and Liu use $\varepsilon_c = 0.10$, five-percent sidebands, and the maximum-growth frequency coordinate $\Delta\omega/(\omega_c\varepsilon_c) = 1$ [17, Sec. 4.3.1]. Under the leading deep-water narrowband relation $\Delta k/k_c \simeq 2\Delta\omega/\omega_c$, this corresponds to $\beta \simeq 1/\sqrt{2}$ in our symmetric-wavenumber normalization. The ten-percent endpoint below is the sideband ratio in the Xu–Guyenne test tuple in [equation \(62\)](#). Revision 4 uses the declared support

$$\begin{aligned} n_c &\in \{4, \dots, 20\}, & \Delta n &\in \{1, \dots, n_c - 1\}, & \varepsilon_c &\in [0.05, 0.13], \\ 0 < \beta < 1, qquad \varepsilon_f &\leq F_0, qquad F_0 = \frac{1 + \sqrt{2}}{10}, \\ \rho &\in [0.05, 0.10], & x_0 &\in [0, L]. \end{aligned} \tag{65}$$

The scalar ε_f is a population-screening proxy, not a maximum physical slope, a wave-breaking criterion, or a regularity theorem.

Let $\mathcal{P}$ be the 66 integer pairs $(n_c, \Delta n)$ for which this support is nonempty. Conditional on a pair in $\mathcal{P}$, set

$$\ell = \frac{\Delta n}{2\sqrt{2}n_c}, \quad \varepsilon_{\min} = \max\{0.05, \ell\}, \quad \varepsilon_{\max} = \min\left\{0.13, \frac{-F_0 + 2\sqrt{F_0^2 + 3\ell^2}}{3}\right\}. \tag{66}$$

The implemented sampler draws $\varepsilon_c \sim \text{Unif}(\varepsilon_{\min}, \varepsilon_{\max}]$, $\rho \sim \text{Unif}[0.05, 0.10)$, and $x_0 \sim \text{Unif}[0, L)$. The upper endpoint in [equation \(66\)](#) is obtained by solving $\varepsilon_f = F_0$ at fixed $(n_c, \Delta n)$. The relative phase $\theta = -\pi/4$ is fixed rather than sampled. The closed intervals in [equation \(65\)](#) denote the closure of the physical support; the half-open endpoint conventions above specify finite-precision sampling and do not change the corresponding continuous uniform laws. The quota rule treats each of the 66 pairs in $\mathcal{P}$ as a parameter category. If a family-split accepted-case quota is C , each pair receives either $\lfloor C/66 \rfloor$ or $\lceil C/66 \rceil$ accepted cases, in the fixed enumerated category order. A rejected attempt is replaced in the same pair category. The pair-conditioned sampler and the general category allocator are both implemented.

At the final split quotas, the 16384 training cases assign 248 or 249 accepted cases to every pair, while the 1024 validation cases and 1024 test cases each assign 15 or 16. Additive training chunks contain the differences between these cumulative balanced allocations. Before bulk generation, the direct revision-4 source gate accepted the first proposal in each of the 66 categories. The completed production dataset subsequently retained 18432 cases from 18455 attempts: 16384 training cases, 1024 validation cases, and 1024 test cases. The 23 rejected attempts retained no rows. A source-bound completion audit independently checked the ordered 66-category taxonomy and quotas, exact proposal replay, transaction and array hashes, the 200 endpoint-pinned, approximately uniform selections from each saved grid, stored depths and water columns, and the declared numerical conditions. Its artifact is `outputs/paper_dataset_bf_revision4_jonswap_revision3_literature_aligned_v1/benjamin_feir_completion_audit.json`, with SHA-256 `c155b35276d0cfef6844f6b0db3e91da3ce15579ad8b0f0ca482f44cc912753c`. Two shared Python modules changed after this calculation and their exact run versions were absent from the Git history. Those two files are now preserved, without being imported by current code, in `reproducibility/source_snapshots/`, in the subdirectory `benjamin_feir_revision4_e16773f/`; the accompanying `SHA256SUMS` file equals the hashes recorded by every one of the six chunk source maps. The completion audit also reconstructs the attempted, accepted, and rejected count in every category from the immutable transactions and physically rehashes the other 17 recorded generation sources against the current repository. It remains the binding record for the dataset arrays and transactions. These counts measure the implemented population and numerical path; they do not turn the proxy bound into a physical wave-breaking criterion.

The former common adapter persisted the pair, continuous parameters, translation, and fixed relative phase before constructing the equation-(33) state. In the reduced accepted-quota calculation, the Benjamin–Feir attempt was accepted, committed three rows, and was finite when read by the training loader. This historical transaction used the former paired validation path and exercised proposal, construction, post-acceptance time selection, whole-case commit, and split-aware view construction. It checks order and storage logic only; the full-horizon panel below is the separate method-level numerical test.

For each carrier, define the linear carrier period and the requested horizon by

$$T_c = \frac{2\pi}{\sqrt{gk_c}}, \quad H_{\text{BF}} = 100T_c, \quad T_{\text{BF}} = 0.08 \left\lfloor \frac{H_{\text{BF}}}{0.08} \right\rfloor. \quad (67)$$

The intended horizon is $H_{\text{BF}} = 100T_c$, and the calculation ends at T_{BF} , the last saved-grid time not exceeding that horizon. This follows the carrier-period time coordinate used by Xu and Guyenne while placing every endpoint on the common saved-time grid. Across the 66 equally weighted pair categories, T_{BF} ranges from 140.48 to 314.08 and has mean 174.50; equivalently, $T_{\text{BF}}/T_c \in [99.9541, 99.9973]$.

Resolved-band JONSWAP/TMA random seas. JONSWAP denotes the Joint North Sea Wave Project frequency spectrum, and TMA denotes its standard Texel–Marsen–Arsloe finite-depth modification. Let $h > 0$ be depth, let $k_p > 0$ be the reference peak wavenumber, let $\gamma \geq 1$ be the peak-enhancement factor, and let $g > 0$ be gravitational acceleration. For wavenumber $k > 0$, define

$$\omega(k, h) = \sqrt{gk \tanh(kh)}, \quad c_g(k, h) = \frac{\partial \omega}{\partial k} = \frac{g[\tanh(kh) + kh \operatorname{sech}^2(kh)]}{2\omega(k, h)}. \quad (68)$$

Set $\omega_p = \omega(k_p, h)$ and define the peak period $T_p = 2\pi/\omega_p$. The intended terminal time, saved-time spacing, final saved index, and realized terminal time are, respectively,

$$H = 16T_p = 16 \frac{2\pi}{\omega_p}, \quad \delta t_s = 0.08, \quad J = \left\lfloor \frac{H}{\delta t_s} \right\rfloor, \quad T_s = J\delta t_s = 0.08 \left\lfloor \frac{H}{0.08} \right\rfloor. \quad (69)$$

Thus T_s is the last point on the saved-time grid that does not exceed the intended horizon H . For angular frequency $\omega > 0$, define

$$\sigma(\omega) = \begin{cases} 0.07, & \omega \leq \omega_p, \\ 0.09, & \omega > \omega_p, \end{cases} \quad (70)$$

and the unnormalized JONSWAP density [12]:

$$\tilde{S}_J(\omega) = g^2 \omega^{-5} \exp \left[-\frac{5}{4} \left(\frac{\omega_p}{\omega} \right)^4 \right] \gamma^{\exp[-(\omega - \omega_p)^2 / (2\sigma(\omega)^2 \omega_p^2)]}. \quad (71)$$

For the dimensionless number $z = kh$, define the TMA multiplier [3]:

$$\Phi_{\text{TMA}}(z) = \frac{\tanh^2 z}{1 + 2z / \sinh(2z)}. \quad (72)$$

The cited works supply these spectral forms. Published numerical JONSWAP realizations have used the interval $0.5f_p \leq f \leq 2.5f_p$ [7], and an earlier random-wave calculation used the upper endpoint $\omega_{\text{max}} = 2.5\omega_p$ [11]. We adopt this finite component interval as a reproducible numerical definition, not as a wave-breaking criterion. The resolved Fourier cells, direction split, phase sampling, and

parameter strata below remain choices of the present dataset. The finite-depth frequency density evaluated at wavenumber k is $\tilde{S}_J(\omega(k, h))\Phi_{\text{TMA}}(kh)$. The factor $c_g(k, h) = \partial\omega/\partial k$ in [equation \(68\)](#) converts this frequency density to a wavenumber density.

Let $\Delta k = 2\pi/L$, let $k_j = j\Delta k$ for positive integers j , and let $\mathcal{J} = \{j \in \mathbb{N} : k_j \leq K\}$ with $K = 128$. Revision 4 retains the fixed relative-frequency interval

$$I_p(k) = \begin{cases} 1, & \frac{1}{2} \leq \frac{\omega(k, h)}{\omega_p} \leq \frac{5}{2}, \\ 0, & \text{otherwise.} \end{cases} \quad (73)$$

This is a sharp restriction in frequency relative to the prescribed peak; it is not a cosine taper in absolute wavenumber. The support requires

$$\omega(K, h) \geq \frac{5}{2} \omega_p, \quad K = 128, \quad (74)$$

so the complete upper end of the declared interval lies in the delivered band for every proposed case. For $j \in \mathcal{J}$, define the Fourier-cell endpoints $a_j = \max\{0, k_j - \Delta k/2\}$ and $b_j = \min\{K, k_j + \Delta k/2\}$. Define the cell energy E_j and normalized energy fraction ϖ_j by

$$E_j = \int_{a_j}^{b_j} \tilde{S}_J(\omega(k, h))\Phi_{\text{TMA}}(kh)c_g(k, h)I_p(k) \, dk, \\ \varpi_j = \frac{E_j}{\sum_{\ell \in \mathcal{J}} E_\ell}. \quad (75)$$

Thus $\varpi_j \geq 0$ and $\sum_{j \in \mathcal{J}} \varpi_j = 1$. The implementation evaluates each integral E_j by 16-point Gauss–Legendre quadrature.

Let $H_s > 0$ be the ensemble significant wave height, define $m_0 = (H_s/4)^2$, and let $r_d \in [0, 1]$ be the fraction of the linear wave energy assigned to right-moving modes. For every $j \in \mathcal{J}$, define

$$A_{j,+} = \sqrt{2r_d m_0 \varpi_j}, \quad A_{j,-} = \sqrt{2(1 - r_d) m_0 \varpi_j}. \quad (76)$$

The phase arrays $(\phi_{j,+})_{j \in \mathcal{J}}$ and $(\phi_{j,-})_{j \in \mathcal{J}}$ are independent, and every entry is uniform on $[0, 2\pi)$. Let $\mathcal{G}_0(h)$ be the flat-surface DNO, whose symbol at positive wavenumber k is $\mathcal{G}_0(k; h) = k \tanh(kh)$. The initial state is

$$\eta_0(x) = \sum_{j \in \mathcal{J}} [A_{j,+} \cos(k_j x + \phi_{j,+}) + A_{j,-} \cos(k_j x + \phi_{j,-})], \\ \xi_0(x) = \sum_{j \in \mathcal{J}} \frac{\omega(k_j, h)}{\mathcal{G}_0(k_j; h)} [A_{j,+} \sin(k_j x + \phi_{j,+}) - A_{j,-} \sin(k_j x + \phi_{j,-})]. \quad (77)$$

The minus sign in the left-moving part of ξ_0 is the linear traveling-wave relation. The construction fixes $\frac{1}{2} \int_0^L (\xi_0 \mathcal{G}_0(h) \xi_0 + g \eta_0^2) \, dx = g L m_0$ independently of phase; it does not rescale an individual realization to force its sampled surface variance to equal m_0 .

For every stratum, define the input peak steepness

$$\varepsilon_p = \frac{k_p H_s}{2}. \quad (78)$$

The declared support has three depth strata. For the shallow stratum, let $n_p \in \{16, 18, 20, 22, 24\}$, set $k_p = 2\pi n_p/L$, and define $\chi_p = k_p h$ and $\delta_s = H_s/(2h)$. Its support is

$$\chi_p \in [0.2, 1.5], \quad h = \chi_p/k_p, \quad \delta_s \in [0.03, 0.16], \quad \varepsilon_p = \chi_p \delta_s \leq 0.08. \quad (79)$$

The finite-depth and deep supports are, respectively,

$$k_p \in [2, 12], \quad h \in [0.1, 1.5], \quad H_s \in [0.005, 0.03], \quad \varepsilon_p \leq 0.08, \quad (80)$$

$$k_p \in [2, 12], \quad h \in [5, 25], \quad H_s \in [0.005, 0.03], \quad \varepsilon_p \leq 0.08. \quad (81)$$

Every depth stratum also satisfies the resolved-frequency condition [equation \(74\)](#). Every stratum uses $\gamma \in \{1, 3.3, 5\}$ and $r_d \in \{0, \frac{1}{2}, 1\}$. The parameter categories are the 27 combinations of depth stratum, γ , and r_d , with requested accepted-case counts divided across those categories by quotient and remainder.

The proposal law inside a shallow category is the following. Draw n_p uniformly from the five values

$$n_p \in \{16, 18, 20, 22, 24\}.$$

Draw χ_p uniformly on $[0.2, 1.5]$ and δ_s uniformly on $[0.03, 0.16]$, repeating the pair until both $\chi_p \delta_s \leq 0.08$ and [equation \(74\)](#) hold. This is the uniform distribution, with respect to area, on the part of the displayed rectangle that satisfies both conditions. Then set $k_p = 2\pi n_p / L$, $h = \chi_p / k_p$, and $H_s = 2h\delta_s$. Inside a finite-depth or deep category, draw k_p , h , and H_s independently and uniformly over the intervals in [equation \(80\)](#) and [equation \(81\)](#), repeating the triple until $k_p H_s / 2 \leq 0.08$ and [equation \(74\)](#) both hold. Equivalently, the resulting proposed triple is uniform on the part of the corresponding rectangular box that satisfies both displayed conditions. In every category, all entries of both phase arrays are independent and uniform on $[0, 2\pi)$.

The bound $\varepsilon_p \leq 0.08$ is a restriction on the population being studied, not a theorem that waves break immediately above it. Ducroz et al. describe $k_p H_s / 2 = 0.08$ as typical of a moderate JONSWAP sea state in a HOS calculation [8]. A separate applicability study shows that increasing the resolved HOS bandwidth can expose short-wave content that a coarser calculation suppresses, thereby reducing the range in which an apparently robust computation is accurate [9]. Since the present potential-flow solver has no physical wave-breaking model, we use the moderate value as a conservative input restriction. It does not replace the full-trajectory numerical checks below: random phases and nonlinear focusing can still make a proposed trajectory fail.

The spectrum, phase sampler, support predicate, state [equation \(77\)](#), and 27-category proposal law are implemented. Both realized phase arrays are held during construction but are not persisted; they can be regenerated from the split, family, and attempt seed. The adapter then checks the realized graph condition

$$\min_{\ell} \{h + \eta_0(x_\ell)\} > 0. \quad (82)$$

This condition cannot be decided from $(h, H_s, k_p, \gamma, r_d)$ alone because the minimum depends on the sampled phases. If one realization fails, that case contributes zero rows and is replaced in the same parameter category; other cases in the same proposed numerical batch continue. No deterministic sum-of-amplitudes bound is used to reject otherwise valid phase realizations.

For JONSWAP/TMA, the event A_f in [equation \(28\)](#) specifically includes the phase-dependent initial construction and graph condition above, the nonlinear adjustment, and the autonomous trajectory decision. This is not a separate phase-valued support gate: it is the common complete-case conditioning specialized to this family. The JONSWAP/TMA completion audit reports its effect separately; the analogous common-law statement does not claim measured bias in another family. No retrospective parameter cutoff or bulk rejection-rate gate is applied.

The current adapter does not persist proposal values or construction metrics. It connects a valid construction to the dedicated adjusted trajectory executor, 16-time selection, accepted-row storage, and split-aware view. An earlier reduced accepted-quota calculation used the former paired validation path: its random-sea attempt was accepted, committed three finite rows, and was

finite when read by the training loader. Its intended horizon was $H = 30.0341$, and its realized terminal time was $T_s = 30.0$, in agreement with [equation \(69\)](#). This is historical software-path evidence, not a test of the current exact-contract adjustment and autonomous path, the population law, or a population acceptance rate. The historical pre-adjustment 27-case pilot contains nine deterministic support-spanning cases per depth stratum and uses each (γ, r_d) pair once per stratum; it was not drawn from the population law and remains only a historical validation panel. It does not test the later adjustment and autonomous numerical contract, nor does it test the revision-4 relative-frequency population.

Revision-4 JONSWAP/TMA generation does not insert the linear random sea directly into the autonomous nonlinear equations. Write the nonlinear evolution equation as $z_t = Lz + R(z)$, where $z = (\eta, \xi)$, Lz is its linear gravity-wave part, and $R(z)$ is the nonlinear remainder. During a Dommermuth adjustment [\[5\]](#), the program instead solves

$$z_t = Lz + A(t)R(z), \quad A(t) = 1 - \exp\left[-\left(\frac{t}{10T_p}\right)^4\right]. \quad (83)$$

Dommermuth supplies the ramped-nonlinearity construction; the exponent four, the scale $10T_p$, and the $20T_p$ handoff horizon are fixed project settings validated below rather than unique prescriptions of that method. For each case, this nonautonomous calculation ends at

$$T_{\text{adj}} = 0.08 \left\lfloor \frac{20T_p}{0.08} \right\rfloor, \quad (84)$$

the last saved-grid time not exceeding $20T_p$. Its complete internal-band endpoint is passed to the autonomous solver without projection to P_{128} or componentwise rescaling. The autonomous clock is then restarted at $t = 0$, A is fixed to one, and the production interval has length T_s from [equation \(69\)](#). Since [equation \(83\)](#) depends explicitly on time, Hamiltonian conservation is not imposed during adjustment. Finiteness, positive water depth, and convergence of every implicit stage are still required.

Production rollout and whole-case acceptance. Tanaka cases are integrated over $0 \leq t \leq 200$. A Benjamin–Feir case is integrated over $0 \leq t \leq T_{\text{BF}}$, with T_{BF} defined in [equation \(67\)](#); a random-sea autonomous case is integrated over $0 \leq t \leq T_s$, with T_s defined in [equation \(69\)](#). Tanaka revision 3 uses the order-six DNO on 1024 points in its filtered internal band $K_{\text{evol}} = 256$, as defined above. Benjamin–Feir revision 4 uses the order-four DNO on 1024 points in the sharp band $|k| \leq 256$. JONSWAP/TMA revision 4 uses the order-four DNO on 2048 points in the sharp band $|k| \leq 704$. Both sharp-band methods use padding factor eight and sharply project both state fields to P_{256} or P_{704} , respectively, after every completed GL2 step. This repeated projection enforces the stated Galerkin space but is not a Hou–Li filter: it applies no smooth damping to retained modes. Their constructed P_{128} states are embedded in the corresponding internal grid and band; for JONSWAP/TMA the full-band result of [equation \(83\)](#) replaces that initial state before autonomous evolution. In every family the delivered state is a 1024-point $(P_{128}\eta, P_{128}\xi)$ pair, and its label remains the order-six quantity q_{ref} in [equation \(29\)](#). Thus the order-four operator determines the Benjamin–Feir and random-sea dynamics, whereas the order-six operator defines the learning label.

The production integrator uses GL2 step 0.01 and saves candidate states every 0.08, without temporal interpolation. JONSWAP/TMA permits at most five fixed-point updates per implicit stage; the other trajectory contracts retain their declared cap of four. An attempted case must first satisfy the parameter and construction support declared for its family. Conditional on that support, a case is retained exactly when its declared numerical path reaches the requested terminal time,

every implicit stage is marked converged, the stage and updated state are finite, every relative stage residual is finite and at most 10^{-8} , all saved η , ξ , and q_{ref} values are finite, and

$$\min_{j,\ell} \{h + \eta(x_\ell, t_j)\} > 0.$$

Revision-4 Benjamin–Feir and JONSWAP/TMA cases have one further numerical condition on their autonomous saved frames. Let $K_F = 256$ for Benjamin–Feir and $K_F = 704$ for JONSWAP/TMA, and define

$$H_F(t_j) = \frac{1}{2} \int_0^{2\pi} \left\{ \xi G_4^{K_F}(\eta; h) \xi + g \eta^2 \right\} dx, \quad d_{H,F} = \max_j \frac{|H_F(t_j) - H_F(0)|}{\max\{|H_F(0)|, \varepsilon_{64}\}}, \quad (85)$$

where $G_4^{K_F}$ is the order-four DNO evaluated from the full sharp internal state and ε_{64} is the smallest positive normal float64 number. The executor requires $d_{H,F} \leq 10^{-3}$. Before projection to P_{128} , it also evaluates at every saved autonomous frame whether the internal state and $G_4^{K_F}(\eta; h)\xi$ are finite and the minimum value of $h + \eta$. These dense saved-frame quantities, including the initial Hamiltonian and maximum drift, are used for the acceptance decision but are not persisted. The Hamiltonian reference is the adjusted endpoint for JONSWAP/TMA; no Hamiltonian gate is applied during its nonautonomous adjustment.

These statements define one requirement: the fixed method must return a complete admissible trajectory. The individual checks determine that decision, but a rejected attempt’s failure reason is not persisted. They are not separately fitted data filters. An incomplete case contributes no rows, including no finite prefix, and its replacement is drawn from the same predeclared parameter category. Sign changes, slope, spectral roughness, target magnitude, visual appearance, and model error do not enter this decision. Hamiltonian drift enters only through the declared autonomous revision-4 Benjamin–Feir and JONSWAP/TMA condition [equation \(85\)](#).

Method-level time-step validation. The production time step was fixed before bulk generation by a separate comparison of two trajectories. For steps Δt_c and $\Delta t_f = \Delta t_c/2$, let $y_{c,j}^{(\alpha)}$ and $y_{f,j}^{(\alpha)}$ denote, at common saved time t_j , the coarse and fine versions of the three fields

$$y^{(1)} = \frac{P_K \eta}{h}, \quad y^{(2)} = \frac{P_K \xi}{h \sqrt{gh}}, \quad y^{(3)} = \frac{q_{\text{ref}}}{\sqrt{gh}}, \quad (86)$$

where $g > 0$ is gravitational acceleration. With $\|v\|_{2,N}^2 = N^{-1} \sum_{\ell=0}^{N-1} |v_\ell|^2$, the refinement defect is

$$E = \max_{\alpha=1,2,3} \frac{\max_j \|y_{c,j}^{(\alpha)} - y_{f,j}^{(\alpha)}\|_{2,N}}{\max_j \|y_{f,j}^{(\alpha)}\|_{2,N} + 10^{-12}}. \quad (87)$$

The validation panel used $\Delta t_c = 0.01$ and $\Delta t_f = 0.005$ and compared E with 10^{-3} . This comparison, and the optional 0.005/0.0025 diagnostic implemented by the audit utility, are not run for individual production cases and do not choose which trajectory is stored.

The two Stokes propagation cases in the predeclared full-horizon panel have defects $E = 5.42 \times 10^{-8}$ (finite depth) and 8.24×10^{-8} (deep water). Five of the seven Tanaka and Benjamin–Feir stress cases pass with $E \leq 1.13 \times 10^{-5}$. The remaining steep seam-centered Tanaka case and equation-(33) stress case become nonfinite at essentially the same physical times under both step sizes. Thus halving the step changed no whole-trajectory decision. Under the production rule, those two incomplete 0.01 trajectories would contribute no rows; their failure is not a large value of the refinement defect. The Tanaka members of this panel used the earlier $K_{\text{evol}} = 128$ contract, before revision 3 introduced the conditional support and buffered evolution described above. They are historical method-development evidence, not a validation of the revision-3 Tanaka contract.

Historical pre-revision-3 failure-mechanism audit. Separate restart experiments diagnosed the two incomplete stress cases under the earlier Tanaka and Benjamin–Feir contracts; they do not alter the dataset rule. The restart times are chosen before the baseline tail exceeds a fixed guard, so we distinguish the source-history measure from the post-restart arm measure. Let $c_k^{\text{src}}(t)$ be the k -th Fourier coefficient of the source baseline DNO field, let $c_k^{\text{r}}(t)$ be that coefficient for one restarted arm, and let t_r be its restart time. In each arm, “DNO field” means $P_K^{\circ}[\mathcal{G}_M^{N,p}(\eta(t); h)\xi(t)]$, evaluated with that arm’s values of N, M, p , and K . For wavenumber bounds $0 \leq a \leq b$ and any coefficient sequence $c = (c_k)$, define

$$\mathcal{E}_{a:b}[c](t) = \sum_{a \leq |k| \leq b} |c_k(t)|^2, \quad [z]_+ = \max\{z, 0\}.$$

Here $\mathcal{E}_{a:b}$ is the energy in wavenumbers $a \leq |k| \leq b$, and $[z]_+$ is the positive part of z . The two growth measures are

$$G_{\text{B}}^{\text{src}}(t) = \frac{[\mathcal{E}_{80:128}[c^{\text{src}}](t) - \mathcal{E}_{80:128}[c^{\text{src}}](0)]_+}{\mathcal{E}_{1:128}[c^{\text{src}}](0)}, \quad (88)$$

$$G_{\text{B}}^{\text{r}}(t) = \frac{[\mathcal{E}_{80:128}[c^{\text{r}}](t) - \mathcal{E}_{80:128}[c^{\text{r}}](t_r)]_+}{\mathcal{E}_{1:128}[c^{\text{r}}](t_r)}. \quad (89)$$

The subscript “B” denotes the terminal band $80 \leq |k| \leq 128$; the superscripts “src” and “r” denote the source trajectory and restarted arm, respectively. At the selected restart, the source measures are $G_{\text{B}}^{\text{src}}(80) = 4.65 \times 10^{-5}$ for Tanaka and $G_{\text{B}}^{\text{src}}(88) = 6.26 \times 10^{-5}$ for Benjamin–Feir, both below the predeclared 10^{-4} guard.

In the one-factor Tanaka restart from $t_r = 80$, the baseline $N = 1024, p = 8$ arm and the $N = 1024, p = 16$ arm both fail at $t = 119.48$, whereas the $N = 2048, p = 8$ arm remains finite with every GL2 stage solved through $t = 122$, but reaches $\max_t G_{\text{B}}^{\text{r}}(t) = 0.562$. The high-band threshold times are nearly unchanged in all three arms. A fourth arm changes only the delivered cutoff from $K = 128$ to $K = 192$. It remains finite with every stage solved and has $\max_t G_{\text{B}}^{\text{r}}(t) = 3.19 \times 10^{-5}$, below 10^{-4} , in the former terminal band $80 \leq |k| \leq 128$, while permitting nonzero energy in $129 \leq |k| \leq 192$. The largest energy in that added band is 2.80×10^{-4} times the restart-time nonzero-mode energy, so the improvement is not caused by a large amount of newly admitted energy. The $M = 5$ and $M = 4$ arms also remain finite with every stage solved through $t = 122$, but reach $\max_t G_{\text{B}}^{\text{r}}(t) = 0.463$ and 0.390 , respectively. Lowering the series order therefore avoids the terminal solve failure in this restart without removing the preceding high-band accumulation.

The corresponding Benjamin–Feir restart gives the opposite cutoff result. The $K = 128, M = 6$ baseline fails at $t = 108.78$, while changing only to $K = 192$ makes the first failed stage earlier, at $t = 103.755$. Changing only to $M = 5$ remains finite with every stage solved through $t = 112$, but still reaches $\max_t G_{\text{B}}^{\text{r}}(t) = 0.182$. The $M = 4$ arm also remains finite with every stage solved through $t = 112$, but reaches $\max_t G_{\text{B}}^{\text{r}}(t) = 0.283$. Over the finite saved prefix of the $K = 192$ arm, its largest energy in $129 \leq |k| \leq 192$, divided by $\mathcal{E}_{1:128}[c^{\text{r}}](t_r)$, is 0.0482 . Thus extra product padding does not remove either failure, a finer base grid prevents only the Tanaka terminal breakdown, and a larger delivered band suppresses the Tanaka accumulation but worsens the Benjamin–Feir failure. Lower series order avoids both terminal failures without suppressing their high-band growth: both lower-order arms cross 10^{-4} , 10^{-3} , and 10^{-2} at $t = 92.32$, 96.8 , and 101.2 , the same times as the $M = 6$ baseline. The restart calculations inherit the baseline history before their restart times and do not justify a change to the common dataset target. Any replacement numerical contract must instead pass a complete-horizon comparison from the original initial condition. The

Tanaka revision-3 boundary panel summarized by [equation \(58\)](#) and [equation \(60\)](#) is that separate comparison; no restarted arm enters the revision-3 support decision.

In the historical pre-adjustment revision-2 panel, the shallow JONSWAP/TMA case reaches the last saved-grid time not exceeding 16 peak periods with $E = 4.52 \times 10^{-6}$ and every implicit stage solved. The finite-depth case also reaches that last saved-grid time, with $E = 5.63 \times 10^{-6}$, all 3864 coarse and 7728 fine stages solved, and no iteration-cap or nonfinite event. The deep case does the same with $E = 4.75 \times 10^{-6}$, all 3496 coarse and 6992 fine stages solved, and no cap or nonfinite event. In total, 10 of the 12 fixed cases complete at both steps: both Stokes cases, three of four Tanaka cases, two of three Benjamin–Feir cases, and all three random-sea cases. No 0.0025 diagnostic was invoked. These are method-validation cases; they do not constitute a generated dataset or estimate population acceptance rates.

Historical JONSWAP/TMA spatial-cutoff validation. Before revision 4 changed the spectral population, the random-sea internal cutoff was selected on a predeclared shallow revision-3 sentinel after its complete nonlinear adjustment. Both arms used 2048 internal grid points, $M = 4$, padding factor eight, GL2 step 0.01, saved spacing 0.08, residual tolerance 10^{-8} , and an iteration cap of five. Over the complete 16-peak-period autonomous history, the largest relative whole-field discrepancies between independently evolved $K = 704$ and $K = 768$ arms were

$$(e_\eta, e_\xi, e_{G\xi}) = (4.64267, 1.93035, 7.91931) \times 10^{-4}. \quad (90)$$

Here each field was transferred by the normalized Fourier construction in [equation \(30\)](#), and the order-six target was recomputed on the common 1024-point grid. The corresponding $K = 640$ versus $K = 768$ calculation exceeded 10^{-3} . Errors computed only from modes $96 \leq |k| \leq 128$ were about 3×10^{-3} , so [equation \(90\)](#) certifies only the three global whole-field quantities, not a bandwise error bound. In particular, this calculation supports the inherited $K = 704$ evolution method but does not validate the revision-4 relative-frequency population.

The then-current revision-3 source was replayed from the archived full internal $K = 704$ endpoint. It reproduced all 289 autonomous saved frames with maximum relative L^2 errors 3.99×10^{-16} , 4.66×10^{-16} , and 1.17×10^{-15} in η , ξ , and the recomputed target. The replay was accepted, solved every stage, had maximum residual 9.994198×10^{-9} , internal Hamiltonian drift 1.831187×10^{-5} , minimum water column 0.0387699, and CPU wall time 197.355 seconds. This verifies that the then-current source reproduces the archived sentinel. It remains historical method evidence.

The direct revision-4 population gate instead fixed 20 accepted cases in each of the 27 categories before execution. It retained all 540 requested cases from 548 attempts. Eight attempts failed the declared complete numerical path and retained zero rows, giving rejection rate 1.46%. This is the current-constructor gate used before revision-4 bulk generation.

3.4 Historical v9 quality controls and the Tanaka generator revision

The historical v9 acceptance tests are source-dependent because the dataset was assembled in stages. We therefore distinguish prevention during construction, rowwise acceptance at assembly, whole-trajectory acceptance for the later nonlinear additions, and an independent grid-refinement audit. The sign-transition screen used for an earlier Tanaka pilot dataset is documented separately in [section A](#); it was not applied to the v9 dataset used by the reported model.

Prevention during construction. The first Tanaka pilot generator embedded a solitary-wave profile in the periodic interval by interpolation with zero values outside the tabulated profile. If a

randomly translated crest lay near an endpoint, this operation cut off a non-negligible tail. The resulting endpoint mismatch has slowly decaying Fourier coefficients, which are amplified by the derivatives in a high-order Craig–Sulem expansion. The replacement Tanaka constructor removes this mechanism before integration. Each tabulated profile is extended by zero outside its finite interval, and the generator sums every periodic image whose support intersects $[0, L)$. It evaluates each image by cubic Hermite interpolation using the tabulated elevation and surface slope, evaluates the sum on an $8N$ grid, and Fourier-restricts it to the N -point grid. The initial state and each rollout-derived target are then projected to $|k| \leq 128$. Thus a profile that crosses $x = 0$ is represented by periodic wrapping rather than by truncation.

Historical row and case acceptance. The common assembly pass for the first ten archived sources retains a snapshot only if

$$\eta, \xi, q^*, h, t \text{ are finite,} \quad \|q^*\|_\infty \leq 1, \quad (91)$$

where q^* denotes that source’s stored target. This pass removed 14 751 of 2 048 000 Benjamin–Feir rows: 14 750 were nonfinite and one finite row violated the target-magnitude bound. Before assembly, the two random-sea archives had respectively 26 and 34 nonfinite rows removed. No row from the two principal Tanaka archives was removed by [equation \(91\)](#).

In historical v9 generation, the later shallow–steep and steep-Tanaka sources use a stronger, case-level test. At stored time t_j and node x_n , let $\eta_{j,n}$, $\xi_{j,n}$, and $q_{j,n}^*$ denote the elevation, potential, and source-specific stored target. Define

$$H_j = \frac{\Delta x}{2} \sum_{n=0}^{N-1} (\xi_{j,n} q_{j,n}^* + g \eta_{j,n}^2), \quad D_H = \max_j \frac{|H_j - H_0|}{|H_0| + 10^{-30}}. \quad (92)$$

A case is accepted only when all stored fields are finite, $\min_{j,n} \eta_{j,n} > -0.9h$, and $D_H \leq \tau_H$. We use $\tau_H = 10^{-5}$ for the three shallow–steep sources and $\tau_H = 10^{-3}$ for the steep-Tanaka addition. The three shallow–steep generation runs rejected 253, 230, and 209 cases, while the two steep-Tanaka shards rejected 985 and 996 cases; generation continued until the target row counts were reached. These are acceptance tests on the stored temporal subsample, rather than a claim that one identical filter was retroactively applied to every source family.

Independent fixed-band resolution audit. Let ϑ denote the latent parameters from which an initial condition is constructed. We construct the same ϑ independently, in float64, on $n_j = 2^j N$ points for $j = 0, 1, 2$; Fourier interpolation of the N -point state is not an independent test. Fix the delivered physical band K and define

$$q_j^K = P_K^\circ \left[\mathcal{G}_6^{n_j, 8}(P_K \eta_{n_j}; h)(P_K \xi_{n_j}) \right]. \quad (93)$$

Let Q_j^K be the vector of consistently normalized Fourier coefficients of q_j^K for $|k| \leq K$. Our three-grid discrepancy is

$$\Delta_K(\vartheta) = \max_{0 \leq r < s \leq 2} \frac{\|Q_r^K - Q_s^K\|_{\ell^2}}{\|Q_s^K\|_{\ell^2} + 10^{-30}}. \quad (94)$$

The additive constant only defines the quotient for a vanishing target; it is negligible for every case in the calibration panel. We declare a relative label accuracy budget $\tau_{\text{ref}} = 10^{-3}$ and require $\Delta_K \leq \tau_{\text{ref}}$ when accepting a generator configuration.

Table 2: Three-grid fixed-band discrepancy on the matched Tanaka calibration. The same decision is obtained at all three tested tolerances.

Construction	Cases	Range of Δ_K
Corrected periodic counterfactual	12	5.86×10^{-6} – 1.10×10^{-5}
Historical interior control	4	1.46×10^{-4} – 1.64×10^{-4}
Historical clipped profile	12	2.33×10^{-2} – 7.08×10^{-1}

A matched Tanaka calibration used $N = 1024$, $2N = 2048$, $4N = 4096$, and the historical delivered band $K = 341$. It contained all 12 reconstructable frame-zero sign rejections, four interior controls, and the same 12 rejected crest specifications rebuilt with the corrected periodic construction. Table 2 shows the resulting separation. Decisions were unchanged when τ_{ref} varied over 3×10^{-4} , 10^{-3} , and 3×10^{-3} . For the matched periodic counterfactuals, the median N – $2N$ label defect was 7.03×10^{-6} and the median $2N$ – $4N$ defect was 2.18×10^{-6} ; the clipped construction gave medians 0.0827 and 0.147, respectively. This is a convergence test for the fixed order-six target on a fixed physical band. It does not test Craig–Sulem order truncation, physical admissibility, or time-integration error, which are addressed separately.

For the existing v9 archive this is a generator-level resolution audit, not a claim that all 7 633 989 snapshots were retrospectively filtered by equation (94). For the replacement dataset, this particular three-grid statistic audits the Tanaka construction. The separate cutoff comparisons for other families have only the scope stated with each panel; in particular, the historical revision-3 JONSWAP/TMA sentinel does not validate the revision-4 input population. None of these method-level comparisons is repeated as a per-candidate state or label filter. Individual rollout cases are governed by the declared-family and whole-trajectory rules above.

Historical pilot cleaning. The April Tanaka pilot dataset was in fact cleaned with a dead-zoned sign-count rule before its 500,000-row training subset was drawn. We retain that fact for reproducibility, but do not present the count as a general numerical validity criterion. Its exact definition, removal count, mechanism, and matched periodization counterfactual are given in section A.

3.5 Sampling, splitting, and normalization

Historical v9 split. The historical v9 archive has no trajectory map used by the training loader. If R denotes its total number of stored rows, the loader applies a fixed-seed random permutation to the integers $0, \dots, R - 1$. The validation and test counts are each $\lfloor 0.1R \rfloor$, and all remaining rows are training rows. The training rows occur first in the permutation, followed by validation and then test. This is a row split: two saved times from the same trajectory can occur in different subsets. That fact must be retained when interpreting the reported v9 holdout error.

Replacement-dataset temporal selection. Temporal selection is applied only after a complete reference trajectory has passed the numerical acceptance calculation. A Tanaka trajectory has candidate times $t_j = 0.08j$, $j = 0, \dots, 2500$. Define

$$S_{T,j} = \sum_{\ell=0}^{N-1} |\partial_x \eta(x_\ell, t_j)|^2, \quad A_{T,j} = \frac{|DS_{T,j}|}{S_{T,j} + 10^{-12}}, \quad (95)$$

where $DS_{T,j} = (S_{T,j+1} - S_{T,j-1})/2$ at interior indices and is the corresponding one-sided difference at the two endpoints. This activity is smoothed by a normalized discrete Gaussian with standard deviation 50 candidate-time indices. Values beyond either endpoint are set equal to the nearest endpoint value during this convolution. Denote the smoothed sequence by $\tilde{A}_{T,j}$. The sampling mass is

$$p_{T,j} = \frac{1 - \alpha_T}{2501} + \alpha_T \frac{\tilde{A}_{T,j}}{\sum_{r=0}^{2500} \tilde{A}_{T,r}}, \quad \alpha_T = 0.5. \quad (96)$$

If the smoothed activity is identically zero, its normalized term in [equation \(96\)](#) is replaced by the uniform mass $1/2501$. A deterministic inverse-CDF rule first computes indices at the midpoint quantiles $(\nu + \frac{1}{2})/200$, where $\nu = 0, \dots, 199$. The first and last indices are fixed to 0 and 2500. Each remaining index is restricted so that enough later integer indices remain available and is then advanced if necessary. This gives exactly 200 strictly increasing indices even when the activity density is concentrated.

For a Benjamin–Feir trajectory, let $t_j = 0.08j$, $j = 0, \dots, J_{\text{BF}}$, where $J_{\text{BF}} = T_{\text{BF}}/0.08$. Retain the 200 indices

$$j_\ell = \left\lfloor \frac{\ell J_{\text{BF}}}{199} + \frac{1}{2} \right\rfloor, \quad \ell = 0, \dots, 199. \quad (97)$$

Thus the realized interval $[0, T_{\text{BF}}]$, which is approximately 100 carrier periods, is sampled approximately uniformly, including both endpoints, without introducing an additional event score.

For a random-sea trajectory, let $t_j = j\delta t_s$, $j = 0, \dots, J$, be the dense saved times, where $\delta t_s = 0.08$ and $J = \lfloor H/\delta t_s \rfloor$ are defined in [equation \(69\)](#). Here $t = 0$ is the adjusted full-band endpoint and the beginning of the autonomous production clock, not the linear random sea before adjustment. For each $\ell = 0, \dots, 15$, retain the index

$$j_\ell = \left\lfloor \frac{\ell J}{15} + \frac{1}{2} \right\rfloor. \quad (98)$$

Thus j_ℓ is the nearest integer to $\ell J/15$, and the retained time is t_{j_ℓ} . In particular, $t_{j_{15}} = T_s$ is the last saved-grid time not exceeding $H = 16T_p$. A static Stokes case retains only $t = 0$. These rules change which accepted states are stored; they do not change the target in [equation \(29\)](#).

Replacement-dataset case split and loader sampling. An attempted case is assigned to training, validation, or test before its initial condition is constructed. The three split roots are 2026072210, 2026072204, and 2026072205, respectively. A rejection remains attached to its assigned split and parameter category, and no saved times from one case may belong to different splits. In each epoch, the training loader uses every accepted training row and globally reshuffles the row indices. Equal accepted- case counts across the four families therefore do not imply equal row weight when the families retain different numbers of times per case. Validation uses every accepted validation row in one fixed deterministic order.

A training case from each of the four families contributes

$$1 + 200 + 200 + 16 = 417$$

stored rows in total. Training is generated in four cumulative stages,

$$C \in \{2048, 4096, 8192, 16384\}$$

accepted cases per family, while validation and test remain fixed at 1024 cases per family. The first stage therefore contains $417(2048 + 1024 + 1024) = 1,708,032$ rows and is a learning-curve

Table 3: Attempted/accepted replacement-dataset cases by family and split.

Family	Training	Validation	Test
Stokes	16384/16384	1024/1024	1024/1024
Tanaka	16384/16384	1024/1024	1024/1024
Benjamin-Feir	16404/16384	1024/1024	1027/1024
JONSWAP/TMA	16644/16384	1040/1024	1042/1024

checkpoint, not the completed dataset. The final target contains $417(16384+1024+1024) = 7,686,144$ rows, which is 52,155 rows (0.68%) larger than the 7,633,989-row historical v9 dataset. The four additive training-stage increments contain 2048, 2048, 4096, and 8192 cases per family, respectively; a completed chunk is never enlarged in place. An increment may itself be divided into disjoint execution shards. Stokes, Tanaka, and Benjamin-Feir use one shard for each of these four increments. To balance the two longer-running JONSWAP/TMA worker lanes, its final 8192-case increment uses the three disjoint intervals $[8192, 12288)$, $[12288, 14336)$, and $[14336, 16384)$.

The split-tagged attempt allocator, atomic completed-batch storage, multi-shard manifest, row map, and loader are implemented and tested as family-independent modules. Each completed batch stores its family and split, its ordered parameter-group identifiers, and any accepted row arrays. A rejected slot is identified by the absence of owned rows. The loader uses the stored split and globally shuffles every accepted training row each epoch. The historical exact-target Stokes calculation described above exercised the former transaction. Under Stokes revision 2, one case from each of its four parameter categories passed this exact calculation, with all four quality-decision masks equal to zero. Its completed production splits contain 16384 training cases and 1024 cases in each of validation and test; these artifacts are retained. In the former reduced accepted-quota workflow, the Tanaka, Benjamin-Feir, and random-sea adapters enforced sampling, durable proposal, then construction. One case from each of these three families continued through the former paired GL2 validation path and loader-facing view, and each produced three finite loaded rows. Those family-revision-1 calculations are historical transaction tests, not tests of the current family-revision executors, numerical validation of the paper target, or evidence about any family population.

The exact Tanaka revision-3 pilot was instead run with the paper contract in two complementary validation-tagged shards covering categories 0–5 and 6–10. All 11 attempted cases were accepted, no case was rejected, and the 11 trajectories produced 2200 rows. These rows are diagnostic pilot output: their dedicated roots are excluded from the replacement-dataset manifest, training, and normalization despite their validation split tags. A later historical Tanaka run completed 4096 cases and retained 200 rows from each, but a constructor audit found 64 mislabeled initial conditions caused by the small-amplitude floor described above. A corrected initial condition changes the nonlinear trajectory and its selected times, so the old rows cannot be repaired algebraically or mixed with corrected rows. The corrected revision-3 release instead draws a fresh deterministic population with 16384 training cases and 1024 cases in each of validation and test. It explicitly does not reuse the historical random specifications.

All four family releases and all four cumulative dataset views are complete. The final release retained 73 728 cases from 74 045 attempts, recorded 317 zero-row rejections, and stored 7 686 144 rows. The family totals (attempted/accepted/rejected/rows) are Stokes revision 2, 18 432/18 432/0/18 432; Tanaka revision 3, 18 432/18 432/0/3 686 400; Benjamin-Feir revision 4, 18 455/18 432/23/3 686 400; and JONSWAP/TMA revision 4, 18 726/18 432/294/294 912. No reported model has been trained on this replacement release.

The declared category marginals also close exactly. Each of the four Stokes categories has

4608/4608 attempted/accepted cases. Across the eleven Tanaka categories, six contain 1675, four contain 1676, and one contains 1678 attempted and accepted cases. Across the 66 Benjamin–Feir categories, 32 contain 278 accepted cases, 18 contain 280, and 16 contain 281; only `n_c_04_delta_n_01` has rejections (304 attempted, 281 accepted), while every other category has attempted=accepted. The 27 JONSWAP/TMA categories contain 22 accepted quotas of 683, three of 682, and two of 680; 16 categories account for all 294 rejections. The immutable document handoff records every named category’s exact attempted, accepted, and rejected counts rather than reducing those JONSWAP/TMA counts to a bulk threshold.

The exact SHA-256 identities of the cumulative combined summaries, indexed by training cases per family, are

2048: 2265cb71270accb3f6d9103e90433b664ca813989dfab63403d8612af22d28c3;

4096: 7feb018ee4d94eab10278917c145262026cccab6169eaaa2d8877f7c7b28c9da;

8192: 43b1eb91198e724c6f32b871c2d0efd0d8355533789919df5897ad588e7e85f6;

16384: 32f764cf865c90892ee0329e48fe992cbc24df2707e05fd1af2ee9cd2b84307c.

The final summary binds the following 26-source manifest and trajectory-map SHA-256 identities:

dbb76ed1cba4d0146f3c9643a73d168d3a9fa97bc27c37fe0fd7ad08554bd734
f26888ad0a2c738405fb1662b9410337e1babfe48f7913805b5243ae0f279cc8.

Historical and replacement-dataset normalization. Let $\mathcal{I}_{v9}$ be the set of all historical v9 row indices. Let η_i and ξ_i denote the two input fields in row i , and let q_i^* denote its stored target under that source’s historical label convention. The reported scale-normalized v9 checkpoints use the dataset-wide absolute maxima

$$s_\eta^{v9} = \max_{i \in \mathcal{I}_{v9}, x} |\eta_i(x)|, \quad s_\xi^{v9} = \max_{i \in \mathcal{I}_{v9}, x} |\xi_i(x)|, \quad s_q^{v9} = \max_{i \in \mathcal{I}_{v9}, x} |q_i^*(x)|. \quad (99)$$

For the replacement dataset, let $\mathcal{I}_{\text{train}}$ contain only rows from accepted cases preassigned to training. For $u \in \{\eta, \xi, q_{\text{ref}}\}$, let $u_i(x)$ denote field u at coordinate x in row i , and define

$$s_u = \max_{i \in \mathcal{I}_{\text{train}}, x} |u_i(x)|, \quad u \in \{\eta, \xi, q_{\text{ref}}\}. \quad (100)$$

For the authenticated release these training-only scales are $s_\eta = 0.15583430230617523$, $s_\xi = 0.09394174814224243$, and $s_{q_{\text{ref}}} = 0.12198150902986526$, computed from 6 832 128 training rows. Their statistics fingerprint is

4e8ec96947bc05cee3b375467499b0ae00a7814f9982614084efd859efe7c3b8.

For one replacement-dataset row, the scale-normalized fields are

$$\tilde{\eta} = \eta/s_\eta, \quad \tilde{\xi} = \xi/s_\xi, \quad \tilde{q}_{\text{ref}} = q_{\text{ref}}/s_{q_{\text{ref}}}. \quad (101)$$

The historical formula has the same form after replacing $(s_\eta, s_\xi, s_{q_{\text{ref}}}, q_{\text{ref}})$ by $(s_\eta^{v9}, s_\xi^{v9}, s_q^{v9}, q^*)$. The replacement-dataset depth coordinate is $d_h = \log(h/h_\star)$, where $h_\star = L/(2\pi)$ is a fixed reference length. Because $L = 2\pi$ in the present calculations, d_h has the same numerical value as the historical input $\log h$. The analytic $\mathcal{G}_0 + \mathcal{G}_1$ paths recover physical η and ξ before applying their Fourier symbols, and predictions are returned to physical target units for rollout. This empirical

normalization is distinct from the exact coordinate dilation derived in [section B](#): it changes neither the period nor the depth.

Several audits distinguish missing data from missing structure. Random and source-stratified scans found no nonfinite base rows, and models that later fail in rollout can still predict the stored target q^* accurately on the corresponding exact truth states. Fine-tuning on states harvested from an older model rollout was actively harmful. Their order-six labels were correct for the harvested states, but those off-manifold inputs already contained high-band contamination created by the older model; the resulting physical high-band response was therefore an undesirable training signal, not a corrupted label. These observations motivate the shape-consistency constraint in [section 5](#): more pointwise samples on the same trajectory manifold do not determine the derivative transverse to that manifold.

4 CS-DNO architecture

4.1 Structural requirements

For fixed (η, h) , the DNO is linear and self-adjoint in ξ and is positive semidefinite modulo constant potentials. Its dependence on η is analytic near the flat surface, and the Craig–Sulem coefficients alternate Fourier multipliers with pointwise multiplication by powers of the surface elevation. CS-DNO imposes linearity and self-adjointness, fixes the first two Taylor coefficients, and restricts the learned remainder to second and higher order:

$$\boxed{\mathcal{G}_\theta(\eta; h)\xi = \mathcal{G}_0(h)\xi + \mathcal{G}_1(\eta; h)\xi + R_\theta(\eta; h)\xi, \quad R_\theta(\eta; h) = O(\eta^2).} \quad (102)$$

The analytic terms are given by [\(9\)](#)–[\(10\)](#); the $\mathcal{G}_1$ Fourier chain is evaluated in float64 and is not cut in wavenumber.

The exact first-order term is important for more than low-amplitude accuracy. Its Fourier kernel between modes k' and k is

$$(kk' - g_0(k; h)g_0(k'; h))\hat{\eta}_{k-k'}. \quad (103)$$

In deep water this vanishes for same-sign k and k' . Earlier fully learned models approximated this cancellation using several blocks and features, but transfer-kernel probes revealed persistent same-sign sidebands. Fixing $\mathcal{G}_1$ analytically and forcing R_θ to begin at $O(\eta^2)$ prevents the learned residual from overwriting the null form.

The corresponding literal component deletion is

$$\mathcal{G}_\theta^{(-1)}(\eta; h)\xi = \mathcal{G}_0(h)\xi + R_\theta^{(-1)}(\eta; h)\xi, \quad R_\theta^{(-1)}(\eta; h) = O(\eta^2). \quad (104)$$

This definition holds the learned parameterization and parameter count fixed; only the analytic summand $\mathcal{G}_1$ is deleted. Since $R_\theta^{(-1)} = O(\eta^2)$, the ablated model has zero first variation at the flat surface and cannot recover the missing $\mathcal{G}_1$ by training. It therefore defines a literal component deletion, not a comparison with a different $\mathcal{G}_0$ -only architecture whose learned residual is allowed to begin at $O(\eta)$.

4.2 Surface features and depth-aware multipliers

The Craig–Sulem recurrence motivates two learned ingredients: Fourier multipliers and spatial multiplication coefficients computed from surface powers. Let $\tilde{\eta} = \eta/s_\eta$, where $s_\eta > 0$ is the fixed training-set normalization scale. The C27 configuration uses `cs_n.polys=3`, so the polynomial

inputs are explicitly $\tilde{\eta}$, $\tilde{\eta}^2$, and $\tilde{\eta}^3$. Together with all four enabled derivative/transform channels, these give the feature stack

$$E_0(\eta) = \left[\tilde{\eta}, \tilde{\eta}^2, \tilde{\eta}^3, \partial_x \tilde{\eta}, \partial_x^2 \tilde{\eta}, |D|^{1/2} \tilde{\eta}, \mathcal{H} \tilde{\eta} \right]. \quad (105)$$

Here $|D|^{1/2}$ and $\mathcal{H}$ are the Fourier multipliers with symbols $|k|^{1/2}$ and $-i \operatorname{sgn}(k)$, respectively. A shared pointwise multilayer perceptron with two bias-free dense layers and GELU activations maps these channels to 320 latent features. Polynomial channels expose the surface powers used by the recurrence, while slope, curvature, half-derivative, and Hilbert-transform channels expose geometric and phase information that a pointwise polynomial cannot recover. Nonlinear mixing allows the learned spatial coefficients to depend on orders beyond degree three; the separate lift below enforces their second-order vanishing at the flat surface. There are no learned spatial convolutions in this trunk; nonlocal coupling is introduced by Fourier multipliers.

For each nonnegative Fourier mode represented on the grid, a small MLP produces a real multiplier from the physical dispersion coordinates

$$m_\theta(k, h) = \text{MLP}_\theta \left(k, \tilde{h}, \tanh(\tilde{h}|k|), |k| \tanh(\tilde{h}|k|) \right), \quad \tilde{h} = \min\{h, 5\}. \quad (106)$$

Thus the shallow-to-deep crossover is supplied explicitly rather than learned from unrelated depth channels. The same clipped depth $\tilde{h}$ is used in the analytic $\mathcal{G}_0$ and $\mathcal{G}_1$ terms. Before depth clipping, the combination $h|k|$ is invariant under the simultaneous dilation in (150): h gains the same factor that $|k|$ loses. This motivates exposing $h|k|$ to the multiplier network. The active model also uses absolute k and h and statistically normalized surface features, so we do not claim that its learned residual is exactly dilation-equivariant.

4.3 Self-adjoint Craig–Sulem blocks

Each learned block is a sum of r multiplier–gain–multiplier sandwiches,

$$R_{\theta,b}(\eta; h)\xi = \sum_{j=1}^r M_{b,j}(h)^* [\Phi_{b,j}(\eta) M_{b,j}(h)\xi]. \quad (107)$$

Here $M_{b,j}$ is diagonal in Fourier space and $\Phi_{b,j}(\eta)$ is a real pointwise gain obtained by projecting the shared surface features. Tying the input and output multipliers makes every branch self-adjoint. Since ξ appears exactly once, the full network is exactly linear in ξ . The gain $\Phi_{b,j}$ is not constrained to be nonnegative, so this factorization does not impose positive semidefiniteness on the learned remainder.

The residual must vanish to first order in η . In the implementation, biases are removed from the residual feature path. If $f_\theta(\eta)$ denotes the resulting latent feature, the block receives $f_\theta(\eta) \tanh(f_\theta(\eta))$. Since $f_\theta(\epsilon\eta) = O(\epsilon)$, this lifted feature is $O(\epsilon^2)$. Consequently $R_\theta(0) = 0$ and $D_\eta R_\theta(0) = 0$ for every parameter value, not only at initialization. Eight blocks with 320 branches each are summed and divided by the square root of the total branch count. The C27 configuration has 1,342,400 trainable parameters, approximately 9.8 times fewer than the generic 13.1-million-parameter FNO baseline.

4.4 Why this factorization matters

For fixed (η, h) , a generic neural operator may learn arbitrary nonlinear dependence on ξ and must rediscover both flat-surface dispersion and the first nonlinear cancellation. CS-DNO restricts learning to the unknown higher-order correction. The architecture is globally nonlocal because its

Fourier multipliers have global physical support, while retaining the multilinear organization of the analytic expansion. In earlier single-seed, three-epoch screens on the v7 dataset, replacing the full feature stack by η alone increased validation loss from 0.00701 to 0.0162, whereas reducing eight blocks to one increased it to 0.00765. These short one-step tests identify the feature stack as more important than block count in that setting; they are not long-rollout or C27 ablations.

5 Training objectives

5.1 Base supervised loss

Let $q^\star = \mathcal{G}^\star(\eta; h)\xi$ denote the stored source-protocol reference target defined in [section 3](#), and let $q^\theta = \mathcal{G}_\theta(\eta; h)\xi$ denote the model value. Thus q represents the first Zakharov vector-field component η_t , not an additional state variable. Let B be the batch size, and let hats denote the one-sided real FFT in the spatial coordinate. Within the modal band below, $q^\star$ is the projected order-six reference value for every source. The accepted C27 model uses the mean per-sample relative one-sided spectral L^2 loss

$$\ell_{2,b} = \frac{\left(\sum_{k=0}^{N/2} |\hat{q}_{b,k}^\theta - \hat{q}_{b,k}^\star|^2\right)^{1/2}}{\left[\max\left\{\sum_{k=0}^{N/2} |\hat{q}_{b,k}^\star|^2, 10^{-12}\right\}\right]^{1/2}}, \quad \mathcal{L}_2 = \frac{1}{B} \sum_{b=1}^B \ell_{2,b}. \quad (108)$$

Here $\tilde{q} = q/s_q$ uses the fixed global target scale supplied to the network. Away from the numerical floor, s_q and the fixed FFT normalization cancel. Equation (108) is the implemented one-sided real-FFT norm: the interior modes are not multiplied by their Parseval multiplicity. This convention does not affect the argument below because all positive interior modes carry the same omitted factor.

The single normalization in (108) allocates accuracy according to absolute target power. To make this precise away from the numerical floor, let

$$Q_b = \sum_j |\hat{q}_{b,j}^\star|^2, \quad \mathcal{A}_b = \{k : \hat{q}_{b,k}^\star \neq 0\}, \quad \mathcal{Z}_b = \{k : \hat{q}_{b,k}^\star = 0\}. \quad (109)$$

For $e_{b,k} = \hat{q}_{b,k}^\theta - \hat{q}_{b,k}^\star$,

$$\ell_{2,b}^2 = \sum_{k \in \mathcal{A}_b} \underbrace{\frac{|\hat{q}_{b,k}^\star|^2}{Q_b}}_{\beta_{b,k}} \left| \frac{e_{b,k}}{\hat{q}_{b,k}^\star} \right|^2 + \frac{1}{Q_b} \sum_{k \in \mathcal{Z}_b} |e_{b,k}|^2. \quad (110)$$

Thus a nonzero mode containing 10^{-4} of the carrier target power contributes only 10^{-4} as much when the two modes have the same fractional error. This is appropriate for total one-step error, but it can hide a weak harmonic or Benjamin–Feir sideband whose phase-rate bias accumulates over a long rollout. The second term in (110) penalizes a coefficient that the model creates where the stored target is exactly zero.

5.2 Active-mode phase-space-normalized complex loss

The quantity called the *mode-balanced loss* in the implementation is more precisely an active-mode relative error in the modal vector field $q = \eta_t$. It does not compare Fourier magnitudes alone. It compares the full complex coefficients, so it retains both radial and angular components of the instantaneous modal motion.

For $1 \leq k \leq K$, with $K = 128$, define

$$g_{0,b,k} = k \tanh(\tilde{h}_b k), \quad \omega_{b,k}^2 = g g_{0,b,k}, \quad \tilde{h}_b = \min\{h_b, 5\}. \quad (111)$$

The depth clipping agrees with the model. For $h \geq 5$, its largest discrepancy from the infinite-depth symbol is $1 - \tanh(5) \simeq 9.1 \times 10^{-5}$ relative to k , attained at $k = 1$. Hats in this subsection denote the forward-normalized real FFT of the physical, denormalized fields. For every training example, set

$$P_{b,k} = |\hat{q}_{b,k}^\star|^2 + \omega_{b,k}^2 |\hat{\eta}_{b,k}|^2, \quad S_b = \max_{1 \leq j \leq K} P_{b,j}. \quad (112)$$

This scale has a direct origin. Suppressing the sample index, the flat-surface dynamics of mode k at the clipped depth are

$$\partial_t \hat{\eta}_k = \hat{q}_k, \quad \partial_t \hat{q}_k = -\omega_k^2 \hat{\eta}_k, \quad \frac{d}{dt} (|\hat{q}_k|^2 + \omega_k^2 |\hat{\eta}_k|^2) = 0. \quad (113)$$

Moreover, if

$$H_k^{\text{lin}} = \frac{1}{2} (g |\hat{\eta}_k|^2 + g_{0,k} |\hat{\xi}_k|^2) \quad (114)$$

denotes the linear Hamiltonian contribution of that mode at the clipped depth, then

$$|\hat{q}_k|^2 + \omega_k^2 |\hat{\eta}_k|^2 = 2g_{0,k} H_k^{\text{lin}}, \quad \hat{q}_k = g_{0,k} \hat{\xi}_k. \quad (115)$$

Consequently, in the linear limit, $P_{b,k}$ is the squared phase-space radius of the corresponding oscillator. In nonlinear data, (112) evaluates that motivated radius formula using the nonlinear target $q^\star$; it is only a reference scale and is not asserted to be a separately conserved nonlinear modal energy. The η contribution also prevents a singular relative error when $\hat{q}_k^\star$ passes through a turning point while the oscillator remains active.

The Hamiltonian relation therefore fixes the units of the denominator only. This is not an energy-matching or energy-conservation penalty: the numerator is the full complex vector-field error $\hat{q}_k^\theta - \hat{q}_k^\star$, and the reference scale $P_{b,k}$ is held fixed during optimization. In particular, coefficients with equal magnitude but different phase are not identified.

The implemented soft activity, relative defect, and robust penalty are

$$a_{b,k} = \frac{P_{b,k}}{P_{b,k} + \alpha S_b + \epsilon_a}, \quad \alpha = 10^{-4}, \quad (116)$$

$$r_{b,k} = \frac{|\hat{q}_{b,k}^\theta - \hat{q}_{b,k}^\star|^2}{P_{b,k} + \beta S_b + \epsilon_a}, \quad \beta = 10^{-6}, \quad (117)$$

$$\rho(r) = 2 \left(\sqrt{1 + \min\{r, 100\}} - 1 \right), \quad \epsilon_a = 10^{-24}. \quad (118)$$

All quantities derived from the reference state are held fixed under differentiation. The loss is

$$\ell_{\text{mode},b} = \frac{\sum_{k=1}^K a_{b,k} \rho(r_{b,k})}{\max\{\sum_{k=1}^K a_{b,k}, \epsilon_a\}}, \quad \mathcal{L}_{\text{mode}} = \frac{1}{B} \sum_{b=1}^B \ell_{\text{mode},b}. \quad (119)$$

For $r \ll 1$, $\rho(r) = r + O(r^2)$. For $P_{b,k} \gg \alpha S_b$, $a_{b,k} \approx 1$, so (119) is approximately the average of $|e_{b,k}|^2 / P_{b,k}$ over reference-active modes rather than the target-power-weighted average in (110). Activity is soft, not binary: a mode with $P_{b,k} = 10^{-4} S_b$ has weight 1/2. Since P is a squared scale, this corresponds roughly to one percent of the strongest modal phase-space radius. Modes below

that level fade continuously, while the smaller denominator floor and the capped pseudo-Huber penalty prevent numerical-noise coefficients or a single outlier from dominating.

The connection to phase accumulation is exact at the level of the vector field. Write a nonzero elevation coefficient as

$$\hat{\eta}_k = A_k e^{i\phi_k}. \quad (120)$$

Because $q = \eta_t$, any candidate value of $\hat{q}_k$ induces

$$\hat{q}_k = e^{i\phi_k} \left(\dot{A}_k + i A_k \dot{\phi}_k \right). \quad (121)$$

At the same fixed training state, the reference and model therefore satisfy

$$|\hat{q}_k^\theta - \hat{q}_k^\star|^2 = (\delta \dot{A}_k)^2 + A_k^2 (\delta \dot{\phi}_k)^2, \quad (122)$$

where $\delta \dot{A}_k = \dot{A}_k^\theta - \dot{A}_k^\star$ and $\delta \dot{\phi}_k = \dot{\phi}_k^\theta - \dot{\phi}_k^\star$. Thus the complex loss measures errors in both modal amplitude-growth rate and modal phase rate without estimating or unwrapping a phase angle. For a progressive linear mode, $|\hat{q}_k^\star| = \omega_k A_k$, and hence

$$r_k \approx \frac{1}{2} \left[\left(\frac{\delta \dot{A}_k}{\omega_k A_k} \right)^2 + \left(\frac{\delta \dot{\phi}_k}{\omega_k} \right)^2 \right]. \quad (123)$$

A traveling-wave speed defect gives $\delta \dot{\phi}_k = -k \delta c$ and $c_k = \omega_k/k$, so its phase component is $r_k \approx \frac{1}{2} (\delta c/c_k)^2$. A persistent bias accumulates as $\delta \phi_k(T) = \int_0^T \delta \dot{\phi}_k(t) dt$ and therefore as a spatial translation. This explains why a small one-step defect can matter in a long rollout.

The loss does not pin an absolute position. Simultaneously translating η , $q^\star$, and q^θ multiplies their Fourier coefficients by the same unit complex number and leaves (119) unchanged. It instead penalizes the erroneous modal velocity that would subsequently create a relative translation. It is also complementary to the localized tangent loss below: the tangent loss retains one source-restricted physical-space projection of $q^\theta - q^\star$, whereas (119) is universal and retains the complete complex error of every reference-active mode.

5.3 Hadamard identity

We suppress the fixed depth argument h in this subsection. Define the vertical and horizontal surface velocities associated with a DNO evaluation by

$$B_\theta = \frac{\mathcal{G}_\theta(\eta)\xi + \eta_x \xi_x}{1 + \eta_x^2}, \quad V_\theta = \xi_x - B_\theta \eta_x. \quad (124)$$

For an exact DNO and any surface perturbation ζ , the Hadamard shape-derivative formula is [14, Theorem 3.20]

$$D_\eta \mathcal{G}(\eta)[\zeta]\xi = -\mathcal{G}(\eta)(\zeta B) - \partial_x(\zeta V). \quad (125)$$

This is precisely the identity that makes the geometric expression for ξ_t equal to the negative η -variation of the kinetic energy

$$K_\theta(\eta, \xi) = \frac{1}{2} \langle \xi, \mathcal{G}_\theta(\eta)\xi \rangle. \quad (126)$$

Hence it is stronger and more local than merely penalizing the drift of the scalar Hamiltonian along one approximate time step.

We approximate the derivative by a finite secant,

$$S_\theta = \frac{\mathcal{G}_\theta(\eta + \epsilon\zeta)\xi - \mathcal{G}_\theta(\eta)\xi}{\epsilon} + \mathcal{G}_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta). \quad (127)$$

The randomized direction is zero mean and supported on $0 < |k| \leq 128$. Its Fourier coefficients are divided by the H^1 weight before RMS normalization, preventing the probe energy from concentrating at the largest modes. Its physical RMS is matched to $\max(\text{rms}(\eta - \bar{\eta}), 10^{-3})$, and ϵ is sampled log-uniformly from $[10^{-3}, 3 \times 10^{-3}]$.

Let $W(k)^2 = 1 + k^2$ and P_K denote projection onto the resolved modes. The normalized consistency loss is

$$\mathcal{L}_{\text{shape}}(\theta) = \mathbb{E}_{(\eta, \xi, h), \zeta} \frac{\|W P_K S_\theta\|_2^2}{\|W P_K [\mathcal{G}_\theta(\eta)(\zeta B_\theta) + \partial_x(\zeta V_\theta)]\|_2^2 + 10^{-12}}. \quad (128)$$

The finite secant avoids the mixed input-parameter higher-order automatic differentiation that would arise when optimizing a Jacobian-vector-product penalty directly.

5.4 Localized translation-tangent loss

A translated solitary wave has the same Hamiltonian as the original profile, so scalar energy control cannot determine its propagation speed. A global projection onto η_x can cancel between counter-propagating or interacting crests. C27 therefore uses a local least-squares projection on the three Tanaka training sources (historical source identifiers 5, 6, and 14). First remove the spatial mean from q^θ and q^* and set $e = (q^\theta - \bar{q}^\theta) - (q^* - \bar{q}^*)$. Let $\tilde{h} = \min\{h, 5\}$, and let $K_{\tilde{h}}$ denote periodic Gaussian smoothing with physical width $\tilde{h}$. Define

$$C_{\tilde{h}} = K_{\tilde{h}} * (\eta_x e), \quad E_{\tilde{h}} = \max\{K_{\tilde{h}} * (\eta_x^2), 0\}, \quad (129)$$

$$\delta c_{\tilde{h}}(x) = -\frac{C_{\tilde{h}}(x)}{E_{\tilde{h}}(x) + 10^{-3} \max_y E_{\tilde{h}}(y) + 10^{-30}}. \quad (130)$$

For the eligible batch indices $\mathcal{I} = \{b : \text{source}_b \in \{5, 6, 14\}, \max_x E_{\tilde{h}_b}(x) > 10^{-20}\}$, the dimensionless loss is

$$\ell_{\text{tan,loc},b} = \frac{\int E_{\tilde{h}_b} |\delta c_{\tilde{h}_b}|^2 / (g \tilde{h}_b) dx}{\int E_{\tilde{h}_b} dx}, \quad \mathcal{L}_{\text{tan,loc}} = \frac{\sum_{b \in \mathcal{I}} \ell_{\text{tan,loc},b}}{\max\{|\mathcal{I}|, 1\}}. \quad (131)$$

The $\sqrt{g \tilde{h}}$ speed scale makes the local error dimensionless without an inverse fitted-speed singularity. Unlike (119), this term discards the component of e orthogonal to the locally smoothed translation direction and is inactive outside the designated solitary-wave sources.

5.5 Combined objective and optimization

Let $\chi_{16}(s) = 1$ when the zero-based optimizer step satisfies $s \equiv 0 \pmod{16}$ and let it be zero otherwise. The C27 per-step training objective is

$$\begin{aligned} \mathcal{L}_{\text{C27}}(\theta; s) &= \mathcal{L}_2 + 6w(s)\mathcal{L}_{\text{mode}} + 10\mathcal{L}_{\text{tan,loc}} + 10^{-2}w(s)\chi_{16}(s)\mathcal{L}_{\text{shape}}, \\ w(s) &= \min\left(\frac{s}{500}, 1\right). \end{aligned} \quad (132)$$

The active-mode term is evaluated on every batch and all wave families. The tangent term is evaluated on every batch but averages only the eligible examples. The shape term is evaluated in

float64 on a random global microbatch of eight examples every 16 optimizer steps. This selective evaluation makes the shape regularizer affordable while sampling many independent states and directions over an epoch.

The coefficient 6 in (132) is empirical: the ablation below tests 6 against 0, not its optimality. The three target-dependent terms use the same reference values in different geometries. The mathematical distinction is that $\mathcal{L}_2$ controls total DNO error, $\mathcal{L}_{\text{mode}}$ approximately equalizes the influence of the complete complex error over reference-active modal phase-space scales, and $\mathcal{L}_{\text{tan,loc}}$ selects a local translation component on particular sources. The Hadamard term is the only one that adds an independent operator-derivative identity. Thus (132) has four algebraic summands: three target-dependent losses and one intermittently evaluated identity penalty. The three-term description used below refers only to the target-dependent part; it does not silently absorb or omit $\mathcal{L}_{\text{shape}}$.

Checkpoint selection uses

$$\mathcal{L}_{\text{val}} = \mathcal{L}_2 + 6\mathcal{L}_{\text{mode}} + 10\mathcal{L}_{\text{tan,loc}}, \quad (133)$$

without a Hadamard secant. The primary C27/C28 deletion comparison uses the final epoch-40 checkpoint in both arms; epoch 40 also minimizes each arm’s own validation objective.

We optimize for 40 epochs with AdamW, global batch size 1024, learning rate 2×10^{-5} , weight decay 10^{-4} , a 500-step learning-rate warmup, and cosine decay. Training is data-parallel over two devices. Because the periodic shape regularizer depends on the exact optimizer counter, checkpoint restoration verifies equality of the full train state across replicas, equality of Adam and schedule counters, and exact agreement between expected and observed regularizer activations. Checkpoint data are made durable before atomic publication of metadata.

6 Numerical results

6.1 Evaluation protocol

Truth and surrogate are compared in a float64 rollout harness using the same integrating-factor Gauss–Legendre method, dealiased Zakharov products, spectral projection, output times, and initial conditions. Only the DNO evaluation is replaced in an unguarded comparison; guarded comparisons additionally apply the conditional spectral viscosity in section 2.4 to the surrogate trajectory. The primary metric is final-time relative surface error

$$e_\eta(T) = \frac{\|\eta_\theta(T) - \eta_\star(T)\|_2}{\|\eta_\star(T)\|_2}. \quad (134)$$

We report its median and 95th percentile together with the numbers of nonfinite and finite-but-divergent trajectories; a finite trajectory is called divergent when $e_\eta(T) > 1$. Truth trajectories with a nonfinite state or relative Hamiltonian drift above 10^{-3} are excluded before scoring. We report separately the number of attempted rollouts and the number with a valid reference trajectory.

The full C16 evaluation contains 32 initial conditions from each of ten panels spanning five physical wave families, for 320 trajectories total. Historical archive names append `g0`, `g1`, or `modal` to some panels. These suffixes mark independent generation shards, not different DNO orders or physical regimes. We therefore call them replicates A–C below. The evaluation uses no runtime cap, hard learned-output cutoff, adaptive damping, or case-specific intervention. Thus its finiteness result does not rely on rollout-time post-processing. The guarded C18 experiment is reported separately because the conditional spectral viscosity of section 2.4 is an intentional numerical intervention. Both experiments otherwise use the same float64 harness and cached truth.

The linear and two Stokes panels originate from a legacy archive with $L_{\text{old}} = 164$. Before rollout we apply the fixed-gravity change of variables derived in [section B](#), with $a = 164/(2\pi)$, and then regenerate the truth trajectory on the canonical $L = 2\pi$ grid. Thus $h_{\text{old}} = 1$ becomes $h^\# \approx 0.0383121$ for the linear and finite-depth Stokes panels, while $h_{\text{old}} = 1000$ becomes $h^\# \approx 38.3121$ for the deep-water Stokes panel. No reported rollout directly deploys the learned operator on a noncanonical period.

6.2 Baseline accuracy and the rollout discrepancy

The original CS-DNO, which learned all nonlinear corrections on top of $\mathcal{G}_0$, reached a best single-step validation loss of 9.28×10^{-4} on the v3 data. A 13.1-million-parameter FNO reached 3.69×10^{-3} under a matched loss and data split, despite twice as many epochs. Thus the Craig–Sulem factorization improved one-step accuracy by a factor of four with approximately 11.6 times fewer parameters.

This advantage did not eliminate long-time failures. To distinguish value error from sensitivity to rollout displacement, write $z_n^\star = (\eta_n^\star, \xi_n^\star)$ for a truth state, $z_n = (\eta_n, \xi_n)$ for the corresponding surrogate state, and $\mathcal{Q}_\theta(z) = \mathcal{G}_\theta(\eta; h)\xi$. Let $\mathcal{Q}^\star$ denote the order-six reference evaluation used by the rollout harness. Define

$$b_n = \mathcal{Q}_\theta(z_n^\star) - \mathcal{Q}^\star(z_n^\star), \quad i_n = \mathcal{Q}_\theta(z_n) - \mathcal{Q}_\theta(z_n^\star), \quad \Gamma_n = \frac{\|i_n\|_2}{\|z_n - z_n^\star\|_2}. \quad (135)$$

These diagnostics use the nondimensional variables of the evaluation and the equal-weight product norm $\|(\delta\eta, \delta\xi)\|_2 = (\|\delta\eta\|_2^2 + \|\delta\xi\|_2^2)^{1/2}$. Thus Γ_n is a within-protocol secant diagnostic, not a coordinate-invariant operator norm. Frame-aligned archives showed median relative DNO errors on the truth states of 1.87×10^{-4} and 1.72×10^{-4} for two eventual Tanaka failures. Near their last finite frames, Γ_n reached 8.11 and 8.42, while $\|i_n\|_2 / \|b_n\|_2$ reached 2.64×10^4 and 2.41×10^4 . At the earlier pre-cascade frames, the induced response was concentrated in $k = 32, \dots, 64$ and subsequently spread through $k = 64, \dots, 128$.

The Hadamard defect separated stable and unstable behavior more clearly than the value error. The order-six reference passed the discrete identity at roughly 10^{-5} or below. Earlier learned models exhibited order-one or larger relative defects on pre-cascade failing states, even while retaining small pointwise operator error. A transfer-kernel probe further identified a spurious same-sign coupling from $k_{\text{in}} = 117$ to $k_{\text{out}} = 115$, the type of first-order interaction canceled exactly by [\(103\)](#).

6.3 Ablations and rejected remedies

Several interventions helped local symptoms without resolving the mechanism. Fourfold time-step refinement changed failure times only weakly and nonmonotonically. Two-times dealiasing of the Zakharov right-hand side was negligible on healthy trajectories but delayed terminal failure by only approximately 0.05–0.08 time units on representative failing states. Hard high-band caps reduced some NaNs but produced bounded, inaccurate trajectories and worsened tail error. Soft high-band penalties and stronger stage-gain penalties did not remove the failing mode.

Rejected GL2 stage-tangent experiment. For completeness, we define the stage-gain penalty used in these earlier experiments. Let $\mathcal{T}_\vartheta$ be one Picard update of the two-stage system [\(18\)](#), where $\vartheta = \theta$ uses the learned DNO and $\vartheta = \star$ uses the order-six reference DNO. Starting from $Y^{(0)} = (v_n, v_n)$, we formed the stopped reference predictor $\bar{Y} = \text{stopgrad}(\mathcal{T}_\star(Y^{(0)}))$. For a small random physical perturbation p , inserted identically into the two stages as $\mathbf{p} = (p, p)$, and finite-secant

length ϵ , define

$$w_{\vartheta,s} = \frac{\mathcal{T}_{\vartheta}(\bar{Y} + \epsilon p)_s - \mathcal{T}_{\vartheta}(\bar{Y})_s}{\epsilon}, \quad s \in \{1, 2\}. \quad (136)$$

The perturbations were normalized in the linear Hamiltonian norm

$$\|(\delta\eta, \delta\xi)\|_{E,h}^2 = g \|\delta\eta\|_2^2 + \langle \delta\xi, \mathcal{G}_0(h)\delta\xi \rangle. \quad (137)$$

Let P_B be the smooth Fourier projector onto the tested band. The corresponding gain of stage s was

$$r_{\vartheta,s} = \frac{\|P_B w_{\vartheta,s}\|_{E,h}}{\|p\|_{E,h} + \epsilon_{\text{norm}}}, \quad (138)$$

and the penalty was the squared hinge

$$\mathcal{L}_{\text{stage-gain}} = \mathbb{E} \left[\frac{1}{2} \sum_{s=1}^2 [r_{\vartheta,s} - r_{\star,s} - (\mu r_{\star,s} + \nu)]_+^2 \right], \quad [a]_+ = \max\{a, 0\}. \quad (139)$$

The expectation averages over training states and random low-to-mid and mid-to-mid perturbations. Each p has unit energy norm, ϵ is log-uniform on $[10^{-6}, 10^{-3}]$, and $\epsilon_{\text{norm}} = 10^{-12}$. A companion response-matching term compared $P_B w_{\vartheta,s}$ directly with $P_B w_{\star,s}$. The decisive C8 and C10 runs used $\mu = \nu = 0$, so any learned gain above the reference gain was penalized. This construction did not assert $r_{\vartheta,s} < 1$ and therefore did not impose contraction of the Picard map; it only limited excess gain relative to the discrete reference map. C8 left the Tanaka nonfinite count at 3/32 and worsened both Tanaka and Benjamin–Feir median errors. Restricting the test to $64 \leq |k| \leq 128$ in C10 reduced the count to 2/32, but retained adverse finite-error tradeoffs. We therefore did not include this penalty in (132).

Fine-tuning on model-harvested pre-cascade states increased failures in both Tanaka and Benjamin–Feir tests; inspection showed that the harvested pack already contained the old model’s high-frequency contamination. A direct residual-only use of the fixed- g variables from (151) was also tested: it supplied the learned residual with η/h , $\xi/h^{3/2}$, $q/\sqrt{h}$, and $h|k|$, while leaving the exact physical $\mathcal{G}_0 + \mathcal{G}_1$ backbone unchanged. On the matched one-percent-data smoke, validation loss worsened from 0.0131081 to 0.0142273; on the shallow Tanaka case-27 probe, relative DNO error worsened from 0.005059 to 0.010280 and relative speed bias from 0.0871% to 0.7394%. We therefore rejected that particular learned-residual parameterization. This result does not contradict the exact continuum identity (150); it shows that imposing the scaling on only one component of a mixed physical/learned model did not improve optimization. These tests did not support time-step refinement, output clipping, or contaminated-state fine-tuning as explanations or remedies for the observed failure.

6.4 C16 full-suite result

The clean C16 replay shares the exact $\mathcal{G}_0 + \mathcal{G}_1 + O(\eta^2)$ decomposition and self-adjoint structural constraints of (102), but uses the earlier 860,928-parameter capacity and a relative H^1 plus sparse-Hadamard objective. It predates the current L^2 , mode, and localized-tangent objective (132). The best validation checkpoint occurs at cumulative epoch 39, with validation loss 6.736928×10^{-3} ; epoch 40 is slightly worse at 6.737164×10^{-3} . The value is not directly comparable with the old v3 loss because the dataset, architecture, normalization path, and auxiliary objective differ. Long-time rollout is the relevant test.

No instance of the previously observed NaN mechanism occurs in the unguarded 320-case C16 suite in table 4. Because C16 changed the analytic backbone, residual structure, shape supervision,

Table 4: Clean C16 epoch-39 full-suite results with no rollout-time intervention. Each panel contains 32 attempted initial conditions. Nonfinite counts use all attempts; divergences and error quantiles use only the truth-valid cases shown in the second column.

Evaluation panel	Truth-valid	Nonfinite	Divergent	Median $e_\eta(T)$	95th percentile
Tanaka, replicate A	32	0/32	1/32	0.08643	0.51111
Tanaka, replicate B	32	0/32	1/32	0.11541	0.66002
Benjamin–Feir, replicate A	32	0/32	0/32	0.01208	0.20452
Benjamin–Feir, replicate B	31	0/32	0/31	0.01773	0.06932
Benjamin–Feir, replicate C	31	0/32	0/31	0.01409	0.15342
Random sea, deep	32	0/32	0/32	0.00283	0.02066
Random sea, finite	32	0/32	0/32	0.00542	0.04091
Linear	20	0/32	0/20	0.02707	0.08647
Stokes, deep	32	0/32	0/32	0.00044	0.00112
Stokes, finite	31	0/32	0/31	0.00343	0.02257
Total	305	0/320	2/305		

capacity, and solver path as a bundle, this observation does not isolate one of those ingredients as the cause. The two remaining threshold crossings are both finite Tanaka trajectories, but they do not have the same cause. Tanaka replicate-A case 27 is principally a translation: its final raw error is 1.167, the crest is displaced by eight grid points, and translation alignment reduces the error to 0.0608. The predicted and true final crest amplitudes are 0.003660 and 0.003707. Its initial translation-speed bias is approximately -0.60% , which accumulates secularly while changing the Hamiltonian only weakly.

The historical two-epoch C18 global-tangent fine-tune used

$$c_b(q) = -\frac{\langle q, \eta_{b,x} \rangle}{\|\eta_{b,x}\|_2^2}, \quad \mathcal{L}_{\text{tan, glob}} = \frac{1}{|\mathcal{I}_{56}|} \sum_{b \in \mathcal{I}_{56}} \left[\frac{c_b(q^\theta) - c_b(q^*)}{\max\{|c_b(q^*)|, 10^{-3}\}} \right]^2, \quad (140)$$

where $\mathcal{I}_{56}$ contains source-5 and source-6 samples. This is a single rigid-wave speed fit and is distinct from the localized C27 term (131). It removes the C16 replicate-A divergence: its unguarded median and 95th percentile become 0.07963 and 0.48205, respectively, with 0/32 divergent trajectories. Benjamin–Feir replicate B is essentially unchanged (median 0.01653, 95th percentile 0.07002, 0/32 divergent). Tanaka replicate B, however, retains one divergence and its 95th percentile worsens from 0.66002 to 0.93169. This failure of transfer motivates a separate analysis rather than a larger weight on the same scalar loss.

6.5 The Tanaka replicate-B case-14 mechanism

Case 14 has depth $h = 0.1776006$ and consists of two nearly equal left-going crests with dimensionless amplitudes 0.12514 and 0.11280, separated by $0.57896 = 3.26h$. This geometry is rare, but the orbit is not absent from the data: its initial η is bit-identical to a source trajectory in the v9 dataset, and 163 of that trajectory’s 200 saved states occur in the row-wise training split.

The failure is neither a pure phase error nor an isolated hallucination of the learned DNO. At $T = 200$, translation alignment leaves relative surface errors 0.506 for C16 and 0.587 for C18. Their maximum slopes are 1.124 and 1.532, compared with 0.0777 for truth, and learned-Hamiltonian drift peaks near 22–23% around $t = 176$. Yet teacher-forced DNO errors remain below 1.05×10^{-3} for C16 and 1.17×10^{-3} for C18 over all 251 truth frames. Evaluating the exact order-six DNO on

Table 5: Production-band amplitude $A(t)$ for Tanaka replicate-B case 14.

	t	C16	C18
	20	2.34×10^{-7}	1.94×10^{-7}
	40	8.99×10^{-7}	7.58×10^{-7}
	120	3.39×10^{-6}	3.04×10^{-6}
	140	1.11×10^{-5}	1.04×10^{-5}
	200	5.30×10^{-5}	7.54×10^{-5}

Table 6: C18 Tanaka results before and after the soliton-selective spectral guard. Each row contains 32 scored initial conditions; all remain finite.

Evaluation panel	Guard	Divergent	Median $e_\eta(T)$	95th percentile
Tanaka, replicate A	off	0/32	0.07963	0.48205
Tanaka, replicate A	on	0/32	0.05065	0.48206
Tanaka, replicate B	off	1/32	0.11134	0.93169
Tanaka, replicate B	on	0/32	0.11135	0.78499

the *predicted* states agrees with the learned output to roughly 1–7% through $t = 184$. Thus the dominant high-frequency DNO response is induced by an already contaminated state; pointwise accuracy on the truth orbit does not control the transverse numerical trajectory.

The precursor is visible in (23). For case 14, $A(0) = 4.5345 \times 10^{-9}$ and the truth remains at this scale. In contrast, the predicted amplitudes evolve as follows.

The former absolute trigger $A > 10^{-4}$ therefore never fired. The corrected threshold in (23) begins its transition near $t = 30$, while the relative surface error is still below 0.01. The C18 scalar tangent loss is also poorly matched to this interaction: approximately 41% of its selected source-5/6 rows contain mixed-direction multicrest states, and 99.2% of the initial squared DNO error for case 14 is orthogonal to the global translation direction. Its initial scalar speed error is only 9.6×10^{-5} relative. These observations explain why improving the global scalar diagnostic does not prevent this transverse cascade.

6.6 Causal guard test and guarded C18 result

As a causal control, applying a relative-growth trigger and fixed $k_{\text{eff}} = 64$ to the original C16 Tanaka replicate-B evaluation changes its divergence count from 1/32 to 0/32, its median from 0.1154 to 0.1046, and its 95th percentile from 0.6600 to 0.5809. For case 14 specifically, the final error decreases from 1.1987 to 0.09835, maximum slope decreases from 1.124 to 0.0790, and final learned-Hamiltonian drift decreases from 7.23% to 1.16%. This intervention changes the outcome without changing the checkpoint, which supports the spectral-cascade diagnosis.

The production experiment combines C18 with the intrinsic selector and depth-scaled guard in section 2.4. Its matched Tanaka results are shown in table 6.

For case 14, the production guard reduces final error from 1.2875 to 0.1573 and maximum slope from 1.532 to 0.0794, close to the truth value 0.0777. This is the first single-checkpoint/integrator configuration with no nonfinite or divergent trajectory in either 32-case Tanaka panel.

Two limitations prevent a stronger claim. First, guarded Tanaka replicate-B case 24 ends at $e_\eta(T) = 0.996$, only narrowly below the declared divergence threshold; zero counted divergences therefore does not imply a comfortable tail margin. Second, although (22) makes the guard exactly

Table 7: Exact C27/C28 deletion ablation. Pooled rollout rows use the same 305 truth-valid cases; the Tanaka replicate-B and pooled Benjamin–Feir rows contain 32 and 94 cases, respectively. Ratios larger than one favor C27.

Quantity	C27: $\lambda_{\text{mode}} = 6$	C28: $\lambda_{\text{mode}} = 0$	C28/C27
Validation $\mathcal{L}_2$	0.00092031	0.00089679	0.974
Raw median	0.002850	0.003263	1.145
Raw 95th percentile	0.047694	0.062459	1.310
Raw mean	0.011408	0.016560	1.452
Aligned 95th percentile	0.026263	0.027802	1.059
Alignment-removable 95th percentile	0.031974	0.050395	1.576
Tanaka replicate-B raw 95th percentile	0.055637	0.229728	4.129
Benjamin–Feir raw 95th percentile	0.041526	0.063113	1.520
Benjamin–Feir aligned 95th percentile	0.031766	0.026753	0.842

inactive for the audited non-Tanaka initial conditions, C18 changes the model parameters. Benjamin–Feir replicate B has been rerun and remains stable, but the other seven non-Tanaka panels have not yet been evaluated with C18. Accordingly, the demonstrated claim is 0/64 divergent Tanaka rollouts, not 0/320 for C18.

6.7 Current C27 result and isolated active-mode-loss ablation

The subsequent C27 model retains the architecture (102), replaces the base H^1 objective by (108), and uses the complete objective (132). Both C27 and C28 were evaluated with the same enabled soliton-selective spectral guard. The selector makes it exactly inactive for all 256 non-Tanaka initial conditions. At the saved output cadence, no C27 Tanaka state reaches the nominal 50%-activation trigger, although the archive cannot exclude activation at an unsaved integrator substep; C28 has not received the same inactivity audit. Finiteness statements in this subsection are therefore conditional on this common guarded protocol.

The C27 final epoch-40 checkpoint was evaluated on 32 initial conditions from each of the ten panels. All saved predicted values of η , ξ , and $\mathcal{G}_\theta(\eta)\xi$ are finite in all 320 attempted rollouts. The reference solver is valid in 305 cases. On those cases, the raw terminal error has median, 95th percentile, mean, and maximum 0.002850, 0.047694, 0.011408, and 0.534156, respectively. The numbers of raw errors above 0.25, 0.50, 0.75, and 1 are 1, 1, 0, and 0.

To separate a displacement from a change of shape, define

$$e_{\text{align}} = \min_{s \in \mathbb{T}_L} \frac{\|\tau_s \eta_\theta(T) - \eta_\star(T)\|_2}{\|\eta_\star(T)\|_2}, \quad e_{\text{trans}} = \sqrt{\max\{e_\eta(T)^2 - e_{\text{align}}^2, 0\}}, \quad (141)$$

where $\tau_s f(x) = f(x - s)$. The shift is obtained by exhaustive periodic cross-correlation followed by continuous Fourier refinement. The second quantity is an alignment-removable diagnostic, not a claim that translation and deformation form globally orthogonal nonlinear coordinates.

C28 is the isolated deletion test for (119). It is a fresh 40-epoch replica of C27 with the same initialization seed, data, architecture, optimizer, base L^2 term, localized tangent term, Hadamard term, and schedule. The serialized configurations differ only in $\lambda_{\text{mode}} : 6 \mapsto 0$. Both comparisons use the final epoch-40 checkpoint and byte-identical reference trajectories. The principal results are shown in table 7.

Removing the active-mode term improves the one-step base validation loss by 2.6%, but worsens the pooled raw 95th percentile by 31.0%, the raw mean by 45.2%, and the alignment-removable

95th percentile by 57.6%. The raw threshold counts above 0.25/0.50/0.75/1 change from 1/1/0/0 to 3/1/1/1. The largest regressions are Tanaka replicate-A case 27 ($0.5342 \mapsto 1.1565$) and Tanaka replicate-B cases 24 ($0.03268 \mapsto 0.45041$) and 29 ($0.00636 \mapsto 0.35173$). In the pooled Benjamin–Feir panel, raw 95th-percentile error increases by 52.0% while translation-aligned error decreases by 15.8%. Because population quantiles need not be attained by the same cases, this sign difference does not show that any particular regression is a pure displacement. It shows only that the raw and alignment-removable population tails worsen while this aligned Benjamin–Feir quantile improves.

Every C28 trajectory remains finite under the shared guard, so (119) is not required for finiteness in this matched guarded benchmark. The isolated conclusion is instead that, under this fixed training recipe, the term materially improves the long-time raw and alignment-removable tail that the lower one-step $\mathcal{L}_2$ value fails to predict. The pattern is consistent with the phase/translation mechanism and the phase-space and modal-rate interpretation in [section 5.2](#); the deletion alone does not prove that weak sidebands are the unique mechanism. It also compares one matched seed and coefficients 6 versus 0, so it does not establish that 6 is optimal.

A separate tangent-off training run, C29, also completed 40 epochs with $\lambda_{\text{tan}} : 10 \mapsto 0$ as the only training-configuration change. A post hoc reconstruction from its archived evaluations gives C29/C27 ratios of 1.526 for pooled raw 95th-percentile error, 1.381 for pooled raw mean error, and 1.329 for aligned 95th-percentile error. The pooled Tanaka and Benjamin–Feir raw 95th-percentile ratios are 2.873 and 1.445, respectively, and threshold counts above 0.25/0.50/0.75/1 change from 1/1/0/0 to 2/1/1/0. All C29 predictions are finite. This comparison is not an exact deletion ablation: the recorded time-integrator source hash changed between the two evaluations, and their reference- q and protocol hashes are not identical. It is therefore secondary evidence for retaining the localized tangent term; a causal estimate requires replaying both checkpoints under one frozen evaluator and one reference cache.

6.8 Reproducibility safeguards

The C16 result is reported from the corrected replay only. An older epoch-34–40 continuation is excluded because a device-to-device replication bug restored the correct optimizer state on one GPU and zero or invalid state on the other. This made the periodic shape loss fire out of phase and yielded a nonphysical distributed optimizer trajectory. The clean replay verifies the full replicated state at startup and after each epoch; throughout the epoch-17–40 replay, the observed and expected shape-loss activation counts agree exactly. Both C18 fine-tuning epochs likewise record 373/373 activations. This distinction is important because the corrupted run happened to produce some superficially favorable metrics, but cannot support a scientific comparison.

7 Discussion

The experiments support two conclusions at different levels of certainty. First, one-step accuracy alone is insufficient for long rollout accuracy. Earlier models had relative DNO errors near 2×10^{-4} on exact trajectory states but developed large secant responses transverse to those states. Including $\mathcal{G}_1$ analytically restores the exact first-order null-form cancellation, and the Hadamard loss constrains the shape derivative that enters the Hamiltonian vector field. The clean C16 model, which introduced these changes as a bundle, produced finite predictions in all 320 unguarded rollouts. Because the analytic backbone, learned remainder, shape loss, capacity, and solver path changed together, this comparison supports the combined construction but does not assign the finiteness result to any one component. In particular, the manuscript does not treat the incomplete literal $\mathcal{G}_1$ deletion or the absence of a matched Hadamard deletion as causal evidence.

Second, the C27/C28 comparison isolates the active-mode complex loss within the stated training recipe. Removing it slightly improves one-step $\mathcal{L}_2$ while worsening the raw and alignment-removable rollout tails. This establishes that the aggregate one-step norm and the long-time translation-sensitive error rank these two checkpoints differently. It does not establish that coefficient 6 is optimal, that every affected case is a translation, or that the active-mode term is necessary for finiteness. The completed C29 training run similarly suggests that deleting the localized tangent term is harmful, but its archived rollout comparison used a changed evaluator source and nonidentical reference- q protocol hashes. We therefore treat C29 as secondary evidence until C27 and C29 are replayed with one frozen evaluator.

The spectral guard must also be kept separate from the learned operator. The C27 and C28 headline statistics are conditional on the common guarded protocol. Its initial-profile selector is exactly inactive on the 256 non-Tanaka test conditions, and no saved C27 Tanaka state reaches the nominal activation threshold, but the saved cadence does not rule out activation at an internal integrator substep. Thus the result is a property of the checkpoint–integrator–guard combination, not an unconditional 320-trajectory stability theorem for $\mathcal{G}_\theta$. The unguarded C16 result is the relevant evidence that finite rollouts can be obtained without this intervention, although C16 retains two finite Tanaka errors above one.

The numerical scope is also finite. The reference is the order-six, $|k| \leq 128$ projected DNO under the stated padding and time-integration protocol, rather than the exact continuum DNO. The suite contains 32 fixed attempts per panel, and error statistics use the 305 cases whose reference trajectories pass the declared validity test. These empirical quantiles are descriptive statistics for five constructed wave families, not confidence bounds for an underlying ocean-wave population. The exact mode deletion uses one initialization seed, and the evaluation families are broader than, but not independent of, the data-generation choices used during model development.

The next comparisons should therefore freeze the evaluator and reference cache before replaying the tangent deletion, then test a matched Hadamard deletion from the selected objective. Final accuracy claims should be repeated on fresh case-level test draws from the paper dataset, with errors reported by family and depth as well as in aggregate. A runtime comparison against the same order-six reference implementation is also needed before making a speed claim; parameter count alone does not determine wall-clock cost.

A Historical Tanaka screening and its replacement

This appendix records a preprocessing rule used for the exploratory April FNO dataset. The final v9 dataset was generated independently and did not inherit this rule. We include the details because they explain exactly which pilot rows were removed and how that decision led to the periodic generator revision described in [section 3](#). That replacement constructor and dataset are complete, but the replacement dataset has not been used for reported training.

A.1 The rule that was applied

For one stored row, let q_j^* denote its stored DNO target at grid node x_j , set $q_N^* = q_0^*$, and form the cyclic forward differences $d_j = q_{j+1}^* - q_j^*$ for $j = 0, \dots, N - 1$. Given a relative dead zone ρ , retain only the signs of differences satisfying

$$|d_j| > \rho \max_{\ell} |d_{\ell}| \tag{142}$$

in their cyclic spatial order. If the remaining signs are $s_1, \dots, s_m$ with $m \geq 1$, define

$$C_\rho(q^\star) = \sum_{i=1}^m \mathbf{1}\{s_i \neq s_{i+1}\}, \quad s_{m+1} = s_1. \quad (143)$$

If no difference survives the dead zone, set $C_\rho(q^\star) = 0$. The cleaner used $\rho = 0.03$ and rejected a row when $C_{0.03}(q^\star) > 10$. It removed 548 360 of 4 261 232 rows (12.87%), leaving a 3 712 872-row archive. The 500,000-row Tanaka set used by the April FNO runs was sampled from this retained archive.

A.2 Why it detected the old generator defect

The rule has a restricted interpretation for the pilot family. For the exact water-wave equations, let $q = \mathcal{G}(\eta; h)\xi = \eta_t$. An exactly translating profile $\eta(x, t) = \bar{\eta}(x - ct)$ satisfies

$$q = -c\bar{\eta}_x, \quad q_x = -c\bar{\eta}_{xx}. \quad (144)$$

When the stored approximation $q^\star$ resolves this physical q , the dead-zoned count $C_\rho(q^\star)$ therefore approximately counts transitions between the principal curvature lobes of a translating profile. A smooth unimodal crest has two inflection points. The one-to-three-crest pilot construction consequently has a nominal count of roughly two to six; the cutoff at ten allowed distortion and crest interaction.

The old builder translated a real-line solitary-wave profile and interpolated it with zero values outside the tabulated support. A crest near a periodic endpoint could therefore have a non-negligible tail cut to zero. The resulting endpoint mismatch has slowly decaying Fourier coefficients, and derivatives in the order-six DNO amplify the associated ringing. This produces additional alternating curvature lobes in $q^\star$, which explains why [equation \(143\)](#) responded to the defect.

The connection can be measured on all 254 reconstructable initial conditions from historical batch zero. Define the normalized discrete endpoint mismatch

$$J_\eta = \frac{|\eta(x_0) - \eta(x_{N-1})|}{\|\eta\|_\infty}. \quad (145)$$

Here $\|\eta\|_\infty = \max_j |\eta(x_j)|$. All 12 initial rows rejected by the sign rule had $J_\eta > 0.05$, with median 0.219, while the 242 retained rows had median 2.50×10^{-6} . For each case, the fraction rejected over its 200 stored times had Spearman correlation -0.701 with the initial distance of the nearest crest from the boundary ($p = 8.4 \times 10^{-39}$).

A construction-level counterfactual isolates the cause. The diagnostic script rebuilt the same 12 rejected crest specifications by summing the central profile and its two adjacent periodic images. Those three images were sufficient for these particular compactly supported profiles because all other images vanished on $[0, L)$. This change from nonperiodic zero extension reduced every sign count from 12–38 to 2–8. The replacement constructor does not hard-code three images; it sums every periodic image whose support intersects $[0, L)$. On the three-grid fixed-band audit of [equation \(94\)](#), their median N – $2N$ DNO discrepancy falls from 0.0827 to 7.03×10^{-6} , and their median $2N$ – $4N$ discrepancy falls from 0.147 to 2.18×10^{-6} . Thus the sign rule was responding to the old embedding operation, not merely to globally rough profiles.

A.3 Why it is not a production validity criterion

The count is neither necessary nor sufficient for fixed-band resolution. At the 10^{-3} two-grid tolerance, all 12 historical frame-zero sign rejections fail the fixed-band resolution test, but so do 29 of the 242

sign-retained cases. Thus every sign-rule rejection failed that test, but the rule found only 12 of its 41 failures. Conversely, independently converged periodic multi-crest and Stokes states can have more than ten physical sign transitions. For example, three corrected periodic Tanaka states with 12, 18, and 20 transitions have fixed-band N – $2N$ discrepancies between 4.90×10^{-6} and 7.17×10^{-6} .

Consequently, the sign count is reported only as a family-specific historical curvature screen. The replacement construction removes its failure mechanism by periodic wrapping and spectral restriction, while the cross-grid fixed-band quantity [equation \(94\)](#) tests the numerical property of interest without imposing a bound on the number of physical extrema.

B Uniform rescaling of the Dirichlet–Neumann operator

This appendix records the exact change of variables used to map legacy data from an original spatial period L_{old} to $L = 2\pi$. Set $a = L_{\text{old}}/L > 0$, so $L_{\text{old}} = aL$. We keep the period in the notation here: $\mathcal{G}^{(L)}(\eta; h)$ denotes the DNO on $\mathbb{T}_L$ at depth h , with the unnormalized-normal convention of [\(4\)](#).

B.1 Covariance of the elliptic boundary-value problem

Define the pullback

$$(\mathcal{P}_a f)(X) = f(aX), \quad \mathcal{P}_a : C_{\text{per}}(\mathbb{T}_{aL}) \longrightarrow C_{\text{per}}(\mathbb{T}_L). \quad (146)$$

Here $C_{\text{per}}(\mathbb{T}_\ell)$ denotes the periodic functions on a one-dimensional torus of period ℓ . Let η and ξ be posed on $\mathbb{T}_{aL}$, let the original depth be h_{old} , and let Φ solve the corresponding harmonic-extension problem. Under

$$x = aX, \quad y = aY, \quad \phi(X, Y) = \Phi(aX, aY), \quad (147)$$

the free surface and depth become

$$\eta^b(X) = a^{-1}(\mathcal{P}_a \eta)(X), \quad h^b = a^{-1}h_{\text{old}}. \quad (148)$$

The chain rule gives

$$\phi_{XX} + \phi_{YY} = a^2(\Phi_{xx} + \Phi_{yy})(aX, aY) = 0, \quad \phi_Y(X, -h^b) = a\Phi_y(aX, -h_{\text{old}}) = 0,$$

and the transformed Dirichlet condition is

$$\phi(X, \eta^b(X)) = (\mathcal{P}_a \xi)(X).$$

Thus ϕ is precisely the harmonic extension associated with $\mathcal{G}^{(L)}(\eta^b; h^b)(\mathcal{P}_a \xi)$. Moreover,

$$\partial_X \eta^b = \mathcal{P}_a \eta_x, \quad \Phi_x(aX, aY) = a^{-1} \phi_X(X, Y), \quad \Phi_y(aX, aY) = a^{-1} \phi_Y(X, Y). \quad (149)$$

Substitution into the unnormalized normal derivative proves

$$\boxed{\mathcal{P}_a \left[\mathcal{G}^{(aL)}(\eta; h_{\text{old}}) \xi \right] = \frac{1}{a} \mathcal{G}^{(L)} \left(\frac{1}{a} \mathcal{P}_a \eta; \frac{h_{\text{old}}}{a} \right) (\mathcal{P}_a \xi).} \quad (150)$$

The factor a^{-1} comes from the derivative, not from a normalization of the normal vector. As a consistency check, when $\eta = 0$, every periodic function f satisfies

$$\begin{aligned} \mathcal{P}_a [|D_x| \tanh(h_{\text{old}} |D_x|) f] &= a^{-1} |D_X| \tanh((h_{\text{old}}/a) |D_X|) (\mathcal{P}_a f), \\ D_x &= -i \partial_x = a^{-1} D_X, \quad D_X = -i \partial_X. \end{aligned}$$

B.2 Fixed-gravity Zakharov scaling

Equation (150) leaves the pulled-back Dirichlet datum $\mathcal{P}_a \xi$ unscaled. To obtain a change of variables for the full fixed- g evolution, the amplitude of ξ and the time coordinate must also be rescaled. Let $q = \mathcal{G}^{(aL)}(\eta; h_{\text{old}})\xi$, set $t = \sqrt{a} \tau$, and define

$$\begin{aligned} \eta^\sharp(X, \tau) &= a^{-1} \eta(aX, \sqrt{a} \tau), & \xi^\sharp(X, \tau) &= a^{-3/2} \xi(aX, \sqrt{a} \tau), \\ q^\sharp(X, \tau) &= a^{-1/2} q(aX, \sqrt{a} \tau), & h^\sharp &= h_{\text{old}}/a. \end{aligned} \tag{151}$$

Linearity in ξ and (150) imply

$$q^\sharp = \mathcal{G}^{(L)}(\eta^\sharp; h^\sharp) \xi^\sharp. \tag{152}$$

The first Zakharov equation follows immediately:

$$\partial_\tau \eta^\sharp = a^{-1/2} \mathcal{P}_a \eta_t = q^\sharp.$$

For the second equation, the required factors are

$$\partial_\tau \xi^\sharp = a^{-1} \mathcal{P}_a \xi_t, \quad \partial_X \eta^\sharp = \mathcal{P}_a \eta_x, \quad \partial_X \xi^\sharp = a^{-1/2} \mathcal{P}_a \xi_x.$$

Consequently every term on the right-hand side of (6) acquires the common factor a^{-1} :

$$\begin{aligned} -g\eta^\sharp &= a^{-1} \mathcal{P}_a(-g\eta), \\ -\frac{1}{2}(\xi_X^\sharp)^2 &= a^{-1} \mathcal{P}_a(-\frac{1}{2}\xi_x^2), \\ \frac{(q^\sharp + \eta_X^\sharp \xi_X^\sharp)^2}{2(1 + (\eta_X^\sharp)^2)} &= a^{-1} \mathcal{P}_a \left[\frac{(q + \eta_x \xi_x)^2}{2(1 + \eta_x^2)} \right]. \end{aligned}$$

Hence $(\eta^\sharp, \xi^\sharp)$ solves the same fixed-gravity Zakharov system on $\mathbb{T}_L$ at depth $h^\sharp$.

B.3 Scope of the identity

The covariance requires a common dilation of the horizontal and vertical coordinates; it does not justify changing period and depth independently. It is also an identity of the continuum DNO and the reference dynamics, not a claim that a learned residual is exactly equivariant. The analytic $\mathcal{G}_0 + \mathcal{G}_1$ backbone inherits the covariance, whereas the learned residual used in the reported model also depends on absolute wavenumber, depth, and dataset-level feature normalization. The rejected residual-only reparameterization discussed in section 6.3 therefore tests a model-design choice and does not test (150) itself.

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